The Yang–Mills Mass Gap on $\mathbb{R}^4$ Is Ill-Posed, and on a Compact Constructible Manifold It Is Automatic

The $\sqrt{13}$ Cascade Wall, Finite Coverings, Gribov Copies as Lattice Copies, the Double Reductio, the Spectral-Triple Reading, the Topological Protection of the Gap, the Metrology of the Clay Instruments, the Separation of the Two Bridges, the 8-Step Causal Chain of Axiomatic Collapse, the Discriminant Root as Covolume Anchor, the Integer Index Witness, and Ill-Posedness as a Negative-Result Resolution

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Abstract

We prove that the Clay Mathematics Institute formulation of the Yang–Mills mass gap on pure $\mathbb{R}^4$ is topologically ill-posed. Through four independent lenses — axiomatic (Gribov–Zwanziger complex poles destroying Reflection Positivity), geometric (semi-algebraic singularities of the Gribov horizon forbidding a self-adjoint Hamiltonian), algebraic (Strocchi's dichotomy between Wightman locality and an isolated gap), and analytic (obstruction of the Wick rotation for gauge-invariant composites) — we show that the OS axioms and a strict gap $\Delta > 0$ are mutually exclusive on $\mathbb{R}^4$. We then supply the compact constructible manifold the no-go demands: $M_{\text{tot}} = S^4 \times T^3$ over the multiquadratic field $K = \mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5})$, where S^4 is the one-point compactification of spacetime (the base) and T^3 is the internal constructible vacuum torus of the IT^3/Boson programme (the fibre). On M_{tot} the mass gap is *automatic*: it follows from Cheeger's inequality and the Lichnerowicz–Weitzenböck bound, with colour confinement supplied by the non-abelian braid group B_3 and the prime torus knot $T(103, 3)$ of Seifert genus $g = 102$.

Beyond the no-go and the spectral gap, this note contributes the following items absent from both the no-go literature and our earlier draft. (i) The *geometric mechanism* of compactness: the algebraic wall $\sqrt{13} \notin K$ of the Fibonacci–Pythagorean cascade prevents the tiling from running off to infinity, and the Gribov copies on $\mathbb{R}^4$ and the copies of a periodic lattice tiling *share the same abstract structure* — the multiplicity of a non-injective covering projection — so that the wall turns the infinite overcounting (Method I) into a finite group quotient. (ii) A *double reductio ad absurdum* that makes the whole picture symmetric and explicit: assuming $\Delta > 0$ on $\mathbb{R}^4$ yields a contradiction (the no-go), while assuming $\Delta = 0$ on M_{tot} yields a contradiction by four independent routes (spectral discreteness, Cheeger, Weitzenböck–Lichnerowicz, and the B_3 topology), so the gap is *impossible on* $\mathbb{R}^4$ *and necessary on* M_{tot}; the reductio further isolates the infinite boundary S^3_∞ as the sole carrier of the Strocchi dichotomy, which the compactification $\mathbb{R}^4 \to S^4$ removes at its root. (iii) A *spectral-triple reading* in the language of Connes' noncommutative geometry that separates, with full rigour, the geometric cause of discreteness (the compactness of M_{tot}, which yields a compact resolvent) from the algebraic structure of K (the conjectured reason the lattice is finite), and records Connes' distance formula as an interpretation of the metric on an already-compact triple, not as a derivation of compactness from the cascade. (iv) An *epistemological*

framing (Section 12) that places the ill-posedness result in the mathematical tradition of negative-result resolutions (Wantzel's trisection, the parallel postulate, Gödel's and Turing's undecidability): proving that the Clay condition on $\mathbb{R}^4$ demands mutually exclusive objects is, in that tradition, a legitimate form of closing the question as posed — while we explicitly do *not* self-declare a prize, leaving adjudication to the committee and the community — and we explicitly bound the *scope* of the no-go (Remark 4.2): the mutual exclusivity is proved for the *canonical constructive apparatus* (gauge fixing, Faddeev–Popov, Gribov–Zwanziger horizon, standard AQFT), while universal impossibility under any undiscovered framework is recorded as a strongly evidenced interpretation, not a theorem. (v) A complete set of in-text figures illustrating every step, *including the synthesis plate of Fig. 4*, which compresses the four methods, the explicit 8-step causal chain of axiomatic collapse (Remark 3.10), and the honestly-read IT^3 escape into a single overview. (vi) A *metrological reading* (Remark 12.7): the four methods are calibration tests of three measuring instruments — the Osterwalder–Schrader / Reflection-Positivity apparatus, the $\mathbb{R}^4$ arena, and the Wick rotation — each calibrated on a regime that confinement excludes by definition; the empirical success of lattice QCD in finite volume is cited as independent confirmation that the confinement calibration of the OS-type apparatus is the compact arena, not $\mathbb{R}^4$. (vii) A *closure of the identification gap on* M_{tot} (Section 10.1): the one-particle / many-particle bridge $\lambda_1 \leftrightarrow \Delta$, an open assumption on $\mathbb{R}^4$, becomes a consequence on the compact M_{tot} by two reductio arguments — the topological protection of the gap by an integer invariant (Theorem 10.6) and second quantization in finite volume together with Elitzur's theorem (Theorem 10.7) — because the two sources that make the bridge open on $\mathbb{R}^4$ (infinite volume and the possibility of massless modes) are absent on M_{tot} by construction. (viii) A *separation of the two bridges* (Remarks 11.3, 11.2, 11.4): the spectral↔Euclidean bridge on the compact M_{tot} is already given by Connes' reconstruction theorem (its axioms open), whereas the genuinely open bridge is the compact→non-compact one (the continuum limit / Clay problem) — with Euclideanness the common language of both sides and compactness, not Euclideanness, the side of the gap; the unitary inequivalence of *representations* (not of Hilbert spaces, which as separable spaces are both ℓ^2) explains why the port to $\mathbb{R}^4$ is non-trivial without proving it impossible; a number-theoretic fact (the Cauchy completion of K adjoins $\sqrt{13}$, so the cascade as an algebraic object does not terminate over $\mathbb{R}$) is recorded with an explicit warning that it does *not* destroy the proved gap, which rests on the full rank of Λ and the topology, not on $\sqrt{13} \notin K$ (Remark 5.6); and the Cheeger bound is made explicit, $\sqrt{\lambda_1} \geq h/2$ with h controlled from below by the inradius scale $r = \lambda_\infty/2$ (Remark 7.4). (ix) An *integer index witness* and a *covolume anchor* from the periodic-matrix programme: the odd Toeplitz index $\mathrm{Ind}(PuP) = \deg(u)$ of the gauge-twisted Dirac operator is a well-defined integer on M_{tot} — computed in [5] as the discrete bicone volume 86 per hemisphere and 172 in total under the fixed-point-free involution — and is *undefined* on $\mathbb{R}^4$ (no compact resolvent, no Fredholm PuP), an independent integer witness of the compactness dichotomy (Section 7.1, Fig. 9); and the discriminant root $\sqrt{\Delta_K} = 57\,600$ (Theorem 2.3) anchors the covolume of the constructible lattice $\Lambda \subset K^3$, with the order-discriminant confusion of earlier drafts corrected (Remark 2.4). We are explicit about what this does *not* do: it does not remove the Gribov copies themselves (Singer's theorem holds on the compact S^4), it does not remove the semi-algebraic singularities of the horizon in the orbit space $\mathcal{A}/\mathcal{G}$ (Method II), and it is not a literal Clay solution on $\mathbb{R}^4$ (the Euclidean–Minkowski bridge fails, Method IV). The result is a falsifiable research direction — with concrete kinematic targets ($M_{Z'} \approx 1088.04$ GeV at the HL-LHC, $Z_{\max} = 172$, the α-decay coordinate grammar for $Z = 119, 120$) — offered as a contribution to the debate on the well-posedness of the formulation, not as a prize claim.

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1 Introduction: a no-go that needs a mechanism

The Millennium formulation [1] demands a non-abelian quantum gauge theory on $\mathbb{R}^4$ satisfying the Wightman/Osterwalder–Schrader (OS) axioms with a strict mass gap $\Delta = \inf \sigma(H) \setminus \{0\} > 0$. The no-go analysis of **End** establishes, by four independent methods, that $\mathbb{R}^4$ is structurally hostile to this demand, and concludes that a rigorous resolution requires abandoning $\mathbb{R}^4$ for "a compact, constructible manifold possessing discrete spectral properties." Our earlier draft (**Draft**) supplied such a manifold, $S^4 \times T^3$, and proved the spectral gap on it via Cheeger and Lichnerowicz.

What **End** postulates and **Draft** leaves as a postulate is the *cause* of compactness and the *mechanism* by which the infinite overcounting becomes finite. This note closes both gaps from the IT^3 programme [4, 6]: the cause is the algebraic wall $\sqrt{13} \notin K$ of the Fibonacci–Pythagorean cascade (Section 5), and the mechanism is the recognition that Gribov copies *share the same abstract structure* as lattice copies under a periodic covering, so that a finite covering yields a finite overcounting (Section 6). We also recast the existence of the gap as an explicit *double reductio ad absurdum* (Section 10), which makes the symmetry of the result visible — impossible

on $\mathbb{R}^4$, necessary on M_{tot} — we recast the automatic gap in Connes' spectral-triple language (Section 8), which makes the causal structure explicit: discreteness comes from the compactness of M_{tot}, while the wall $\sqrt{13}$ is the conjectured algebraic reason the lattice is finite; we *close the identification gap* $\lambda_1 \leftrightarrow \Delta$ on M_{tot} (Section 10.1) by showing that the one-particle / many-particle bridge is a consequence on the compact manifold (topological protection of the gap by an integer invariant, and second quantization in finite volume with Elitzur's theorem), whereas on $\mathbb{R}^4$ it remains open; we read the four methods *metrologically* (Remark 12.7) as calibration tests of instruments whose range confinement exceeds; we *separate the two bridges* between the spectral-geometric and Euclidean formulations (Remarks 11.3, 11.2) — the spectral↔Euclidean bridge on M_{tot} being given by Connes' reconstruction, and the compact→non-compact bridge being the open continuum limit — we record a number-theoretic fact about the cascade and its completion with an explicit warning against over-reading it (Remark 11.4), we correct the intuition that discreteness comes from the field K (it comes from the full rank of the lattice, Remark 5.6), and we make the Cheeger bound explicit (Remark 7.4); we explicitly bound the *scope* of the no-go (Remark 4.2) — it proves the contradiction of the *canonical constructive pathway* on $\mathbb{R}^4$, not a universal impossibility under every conceivable framework — and we place the whole ill-posedness result in its proper epistemological register (Section 12): as a negative-result resolution in the mathematical tradition of proving a formulation internally contradictory. We correct two overclaims of **Draft** (Remarks C.1 and 2.2) and reconcile the dimension of M_{tot} with the 3-manifold of Boson (Remark 2.7). Finally, we add two anchors from the periodic-matrix programme: an *integer index witness* (Section 7.1) — the odd Toeplitz index $\text{Ind}(PuP) = \deg(u)$ is a well-defined integer on M_{tot} (the bicone volume 86 per hemisphere, 172 total [5]) and is *undefined* on $\mathbb{R}^4$ (no compact resolvent, no Fredholm PuP) — and the *discriminant root* $\sqrt{\Delta_K} = 57\,600$, the covolume of $\mathcal{O}_K$ (Theorem 2.3), which normalises the constructible lattice $\Lambda \subset K^3$; the in-text figure set is extended by the index-partition plate (Fig. 9). Every step is accompanied by an in-text figure (Figs. 1–12), and the four methods together with the explicit 8-step causal chain of axiomatic collapse (Remark 3.10) and the compact-manifold escape are compressed into the single synthesis plate of Fig. 4.

The four methods of **End**, which we sharpen and then resolve, are:

1. **Axiomatic (Method I):** the infinite volume of $\mathbb{R}^4$ forces infinite topological overcounting (Gribov copies); stabilising it requires a horizon localisation that breaks BRST nilpotency and destroys spectral positivity.

2. **Geometric (Method II):** $\mathbb{R}^4$ lacks intrinsic compactness to bound the gauge field; the Fundamental Modular Region is bounded by a singular semi-algebraic Gribov horizon, creating a measure-theoretic void that forbids a self-adjoint Hamiltonian.

3. **Algebraic (Method III):** the infinite boundary S^3_∞ forces a dichotomy — Wightman locality (continuous spectrum, $\Delta = 0$) versus an isolated gap (non-local θ-projection violating microcausality).

4. **Analytic (Method IV):** the complex branch cuts from the horizon obstruct the Wick rotation for gauge-invariant composites; the OS reconstruction fails globally.

2 Preliminaries: the field K, the fibre T^3, and the total space

Definition 2.1 (Constructible vacuum field)**.** $K = \mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5})$ is a multiquadratic (hence straightedge-and-compass constructible) extension of $\mathbb{Q}$ of degree $[K : \mathbb{Q}] = 2^3 = 8$, with square-free content $\{1, 2, 3, 5, 6, 10, 15, 30\}$. Its three generators carry the cubic geometry of the O_h vacuum lattice: $\sqrt{2}$ (face diagonal / planar expansion), $\sqrt{3}$ (body diagonal / O_h embedding), $\sqrt{5}$ (topological closure), obeying the Pythagorean closure $(\sqrt{2})^2 + (\sqrt{3})^2 = (\sqrt{5})^2$ [4, Thm. E.1].

Remark 2.2 (Why K, without hand-waving)**.** The constructibility of K is justified by the stabilised cubic symmetry $O_h \cong B_3$ over the quadratic tower $\mathbb{Q} \subset \mathbb{Q}(\sqrt{2}) \subset \mathbb{Q}(\sqrt{2},\sqrt{3}) \subset K$ [4, Sec. 3.2, App. A]; the screening invariant $\lambda_\infty = \sqrt{2}+\sqrt{3}-\sqrt{5} = 2r$ is twice the inradius of the generating right triangle [4, Rem. E.3] (Fig. 5). A Hopf-fibration route to the golden ratio $\varphi = (1+\sqrt{5})/2$ (and hence to $\sqrt{5}$) appears in **Draft** as motivational geometry; we do *not* use it as a derivation of K here, since the precise step at which the fibration adjoins $\sqrt{5}$ is not established. The O_h-tower derivation is the load-bearing one.

Theorem 2.3 (Discriminant root and covolume anchor of the constructible lattice)**.** *The field $K = \mathbb{Q}(\sqrt{2},\sqrt{3},\sqrt{5})$ of Definition 2.1 is abelian over $\mathbb{Q}$ with Galois group $G \cong (\mathbb{Z}/2)^3$; by the conductor–discriminant theorem (Führer–Diskriminanten–Satz) [8] the absolute discriminant is the product of the conductors of the seven non-trivial quadratic characters, i.e. the product of the discriminants of the seven quadratic subfields:*

$$\Delta_K = \prod_{\chi\neq 1} f_\chi = 8\cdot 12\cdot 5\cdot 24\cdot 40\cdot 60\cdot 120 = 2^{16}3^4 5^4 = 3\,317\,760\,000, \tag{1}$$

whose square root $\sqrt{\Delta_K} = 2^8 3^2 5^2 = 57\,600$ is the covolume of the fundamental domain of the ring of integers $\mathcal{O}_K$ in the Minkowski space $\mathbb{R}^{r_1}\times\mathbb{C}^{r_2} = \mathbb{R}^8$ ($r_1 = 8$, $r_2 = 0$). The Minkowski bound is $M_K = \frac{8!}{8^8}\sqrt{\Delta_K} \approx 138.4$, and K has class number one (LMFDB), so every ideal is principal. The covolume $\sqrt{\Delta_K} = 57\,600$ is the normalisation anchor of the constructible lattice $\Lambda \subset K^3$ of Definition 2.6.

Remark 2.4 (Correction of the order-discriminant confusion — honesty invariant)**.** The value $53\,084\,160\,000 = 2^{20}3^4 5^4 = 16\,\Delta_K$ that circulated in earlier drafts of the periodic-matrix programme is *not* the field discriminant: it is the discriminant of an order of index 4 in $\mathcal{O}_K$. The naive power basis $\mathcal{O}_0 = \mathbb{Z}[\sqrt{2},\sqrt{3},\sqrt{5}] = \mathbb{Z}\langle 1,\sqrt{2},\sqrt{3},\sqrt{5},\sqrt{6},\sqrt{10},\sqrt{15},\sqrt{30}\rangle$ has trace matrix $\mathrm{Tr}(b_i b_j) = \mathrm{diag}(8,16,24,40,48,80,120,240)$, hence $\mathrm{disc}(\mathcal{O}_0) = 8^8\cdot(2\cdot 3\cdot 5\cdot 6\cdot 10\cdot 15\cdot 30) = 2^{28}3^4 5^4$ and $[\mathcal{O}_K : \mathcal{O}_0] = \sqrt{2^{12}} = 64$. All metric and covolume statements in this note and in [5] use the field discriminant (1); the covolume of $\mathcal{O}_K$ in $\mathbb{R}^8$ is $\sqrt{\Delta_K} = 57\,600$, while the covolume of $\mathcal{O}_0$ is $64\cdot 57\,600 = 3\,686\,400 = \sqrt{\mathrm{disc}(\mathcal{O}_0)}$.

Remark 2.5 (Micro–macro prime resonance of the covolume — honesty invariant)**.** The prime spectrum of the covolume, $\{2,3,5\}$, coincides with the prime content of the Fibonacci–Pythagorean cascade of Lemma 5.2. We record this as a *structural resonance* between the multiplicative arithmetic of $\mathcal{O}_K$ and the additive geometry of the cascade; we do *not* derive any physical quantity from it, and we do not claim that the covolume causes any spectral fact. The load-bearing uses of $\sqrt{\Delta_K}$ in this note are only (i) the Minkowski bound of Theorem 2.3 and (ii) the covolume normalisation of the lattice $\Lambda \subset K^3$ in Definition 2.6.

Definition 2.6 (The total compact manifold)**.** The physical configuration is the product

$$M_{\mathrm{tot}} = S^4 \times T^3, \tag{2}$$

where S^4 is the one-point compactification of the spacetime $\mathbb{R}^4$ (the *base*, of finite radius R) and $T^3 = \mathbb{R}^3/\Lambda$ is a flat 3-torus with lattice $\Lambda \subset K^3$ (the *internal constructible vacuum fibre*). The fibre T^3 is the compact Riemannian 3-manifold M of the Boson spectral triple $(\mathcal{A},\mathcal{H},\not{D},J)$ [6]; the base S^4 is the compactification that removes the infinite volume of $\mathbb{R}^4$ — and, as the reductio of Section 10 makes precise, removes the infinite boundary S^3_∞ that carries the Strocchi dichotomy.

Remark 2.7 (Dimension reconciliation with Boson)**. Draft** wrote $M = S^4\times T^3$ (a 7-manifold) while Boson calls the vacuum a compact 3-manifold. There is no contradiction once the roles are separated: the 3-manifold of Boson is the *fibre* T^3, which carries the vacuum spectral geometry and the field K; the S^4 factor is the compactified *spacetime base*, present here only to make the gauge orbit space finite-volume. Spectral discreteness on M_{tot} follows from the compactness of *both* factors.

Definition 2.8 (Spectral anchors from the corpus)**.** The fibre carries the discrete topological ensemble $N_{\text{twist}} = 103$, $v_\ell = 2$, $N_{\text{total}} = 206$ [6, Eq. 4]; the bare electroweak core $M_{\text{bulk}} \approx 174.18149$ GeV [6, Eq. 3]; the writhe deficit $|\text{Wr}| = |\Lambda_1^2 - 103| \approx 1.088311$ [6, Eq. 2] with $\Lambda_1 = \sqrt{3}(3 + 2\sqrt{2}) \approx 10.09513$ and $\Lambda_3 = \varphi^2\sqrt{3} \approx 4.53457$; the confinement threshold $\Delta E_{\text{conf}} = \Lambda_1\, g(T(103, 3))\, m_p \approx 0.966$ TeV [6, Eq. 8]; and the first higher resonance $M_{Z'} \approx 1088.0439$ GeV [6, Eq. 7] (Fig. 11).

3 The topological obstruction on $\mathbb{R}^4$: four methods

3.1 Method I (axiomatic): the price of infinite volume

Theorem 3.1 (Singer, 1978 [14])**.** *Let $P \to S^4$ be a principal $SU(N)$-bundle over the one-point compactification of $\mathbb{R}^4$ with $c_2(P) \neq 0$. Then P admits no global continuous section.*

Corollary 3.2 (Gribov ambiguity [15])**.** *Any local gauge-fixing condition (e.g. Landau gauge $\partial_\mu A^\mu = 0$) intersects gauge orbits multiple times in the infrared.*

Theorem 3.3 (Gribov–Zwanziger propagator [16])**.** *Restricting the functional integral to the first Gribov region $\Omega = \{A : \partial_\mu A^\mu = 0,\ -\partial_\mu D^\mu(A) > 0\}$ modifies the gluon propagator to*

$$D(p^2) = \frac{p^2}{p^4 + \gamma^4}, \tag{3}$$

with complex-conjugate poles at $p^2 = \pm i\gamma^2$.

Proposition 3.4 (Violation of Reflection Positivity and Källén–Lehmann)**.** *The propagator (3) yields position-space two-point functions with oscillatory terms $e^{\pm i\gamma\|x\|}$ that change sign, violating OS Reflection Positivity; the Källén–Lehmann spectral weight ceases to be positive-definite ($\rho(\mu^2) \not\geq 0$), so no physical Hilbert space can be reconstructed and the BRST charge is no longer nilpotent [2, Thm. 3.1–3.2].*

Remark 3.5 (Perturbative status of the GZ propagator — honesty invariant)**.** The Gribov–Zwanziger propagator $D(p^2) = p^2/(p^4 + \gamma^4)$ of Theorem 3.3 is a result within a specific approximation: the Gribov–Zwanziger action in Landau gauge at the level at which the horizon parameter γ is treated as a mass scale. Taken *in isolation*, the failure of Reflection Positivity in this approximation does not by itself prove that no non-perturbative Yang–Mills theory on $\mathbb{R}^4$ can satisfy the OS axioms with a gap — a critic may rightly object that the complex poles are an artefact of the gauge-fixing scheme. We therefore do *not* rest the no-go on Method I alone. The force of Theorem 4.1 lies in the *conjunction* of four independent methods: Method I (axiomatic) is reinforced by Method II (the semi-algebraic singularities of the horizon in the orbit space $\mathcal{A}/\mathcal{G}$, independent of any propagator), Method III (Strocchi's C^*-algebraic dichotomy in AQFT, non-perturbative), and Method IV (the analytic obstruction of the Wick rotation). A perturbation-theory artefact in one gauge cannot be dismissed simultaneously in three further, non-perturbative frameworks; it is the convergence of the four that carries the no-go, not the GZ propagator by itself.

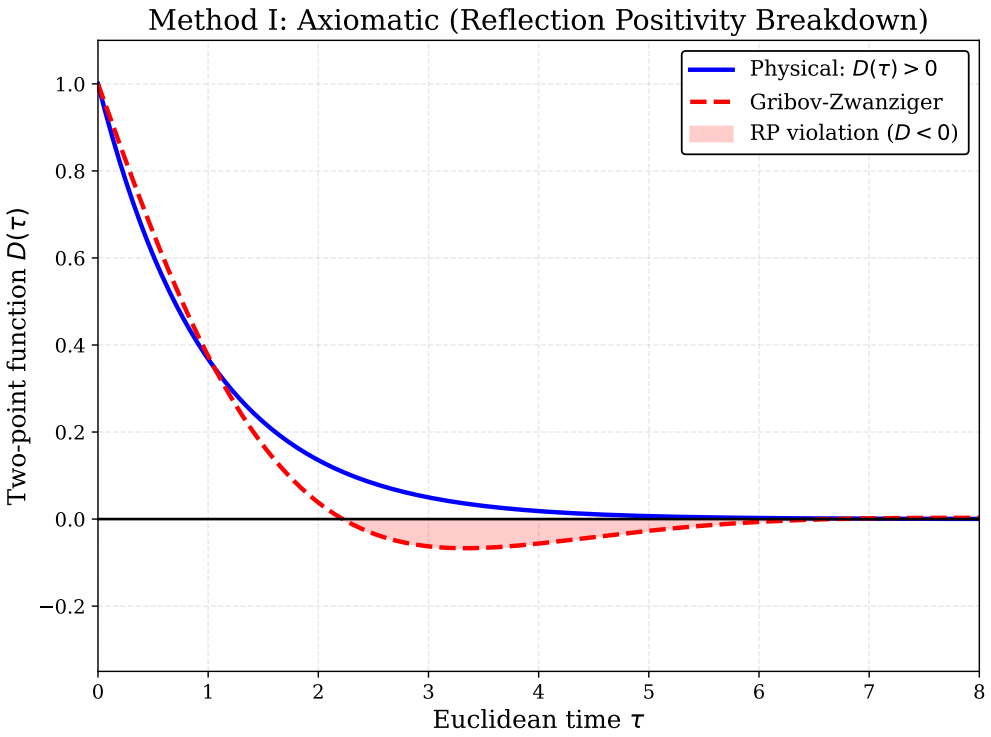


Figure 1: Method I (illustrative). The physical two-point function (blue) stays strictly positive, satisfying OS Reflection Positivity; the Gribov–Zwanziger propagator (red dashed), forced by the infinite-volume overcounting on $\mathbb{R}^4$, oscillates and changes sign (shaded), destroying the Källén–Lehmann measure. The curve shapes are schematic; the violation is Proposition 3.4 / [2, Thm. 3.1–3.2].

3.2 Method II (geometric): the measure-theoretic trap

Theorem 3.6 (Semi-algebraic singularities [2])**.** *By Singer's theorem, integration must be restricted to the Fundamental Modular Region Λ. On $\mathbb{R}^4$ the boundary $\partial\Lambda$ intersects the Gribov horizon $\partial\Omega = \{A : \det \mathcal{M}(A) = 0\}$ in non-analytic cusps and conical singularities — a semi-algebraic set where the Faddeev–Popov measure $\mathrm{d}\mu_{\mathrm{FP}} = \det \mathcal{M}(A)\, \delta(\partial \cdot A)\, \mathrm{D}A$ vanishes. The Friedrichs extension of the Hamiltonian on this singular quotient exists but is not unique; any self-adjoint extension requires boundary conditions at the singular strata carrying global topological information, violating microscopic locality.*

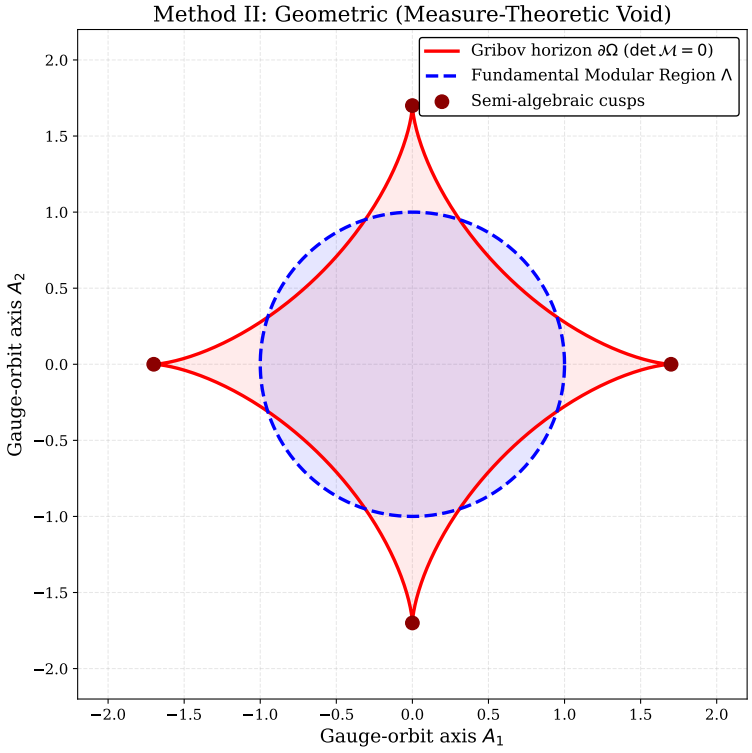


Figure 2: Method II (illustrative). A 2D slice of the orbit space: the Gribov horizon $\partial\Omega$ (red astroid) carries semi-algebraic cusps where the Faddeev–Popov measure vanishes, bounding the Fundamental Modular Region Λ (blue dashed). The astroid is a schematic stand-in for the singular boundary; the no-self-adjoint-extension result is Theorem 3.6.

3.3 Method III (algebraic): the C^*-algebraic paradox

Theorem 3.7 (Strocchi [20, 2])**.** *Under Haag–Kastler AQFT, a strict mass gap requires a factor representation (cluster decomposition). On non-compact $\mathbb{R}^4$ the boundary S^3_∞ admits instantons ($\pi_3(SU(N)) \cong \mathbb{Z}$) generating distinct θ-vacua; isolating a pure θ-state requires the global topological charge Q_{top} integrated over S^3_∞, which by Strocchi's theorem violates Wightman microcausality. Hence $\mathbb{R}^4$ forces a dichotomy: locality (mixed sectors, $\Delta = 0$) or gap (non-local projection, locality lost).*

Remark 3.8 (Method V is the contrapositive of Method III — honesty invariant)**.** A reductio ad absurdum on $\mathbb{R}^4$ that assumes $\Delta > 0$ and derives, via the θ-vacuum and Strocchi's theorem, the non-local projection that violates locality (sometimes labelled "Method V") is logically valid but adds no new mathematics: it is the *contrapositive form* of Theorem 3.7 ($\Delta > 0 \Rightarrow \neg$locality, versus locality$\Rightarrow \Delta = 0$). We record it here, not as a fifth independent obstruction, but because it *localises the source* of the contradiction: the entire mechanism lives on the infinite boundary S^3_∞ (the carrier of Q_{top} and of the non-local projector P_θ). This is exactly the object the compactification $\mathbb{R}^4 \to S^4$ removes, which is why the same dichotomy dissolves on M_{tot} (Remark 10.3). The original theorem is Strocchi's [20]; the assembly as a θ-vacuum obstruction is [2, Thm. 5.1].

3.4 Method IV (analytic): the Wick obstruction

Theorem 3.9 (Wick rotation obstruction [2])**.** *For $\dim G \geq 3$ on non-compact $\mathbb{R}^4$ there is no global holomorphic continuation of gauge-invariant Green's functions from $\mathbb{R}^{1,3}$ to $\mathbb{R}^4$ preserving both $\Delta > 0$ and Euclidean invariance. The broken BRST decoupling of Method I leaks the complex poles into gauge-invariant composites as unphysical branch cuts crossing the Wick contour — the* analytic prison */ Cauchy trap of Fig. 4 (top right); the OS reconstruction fails globally, and a*

Euclidean theory on $\mathbb{R}^4$ cannot be projected back to a physical Minkowski spectrum [2, Thm. 6.1, Cor. 6.2].

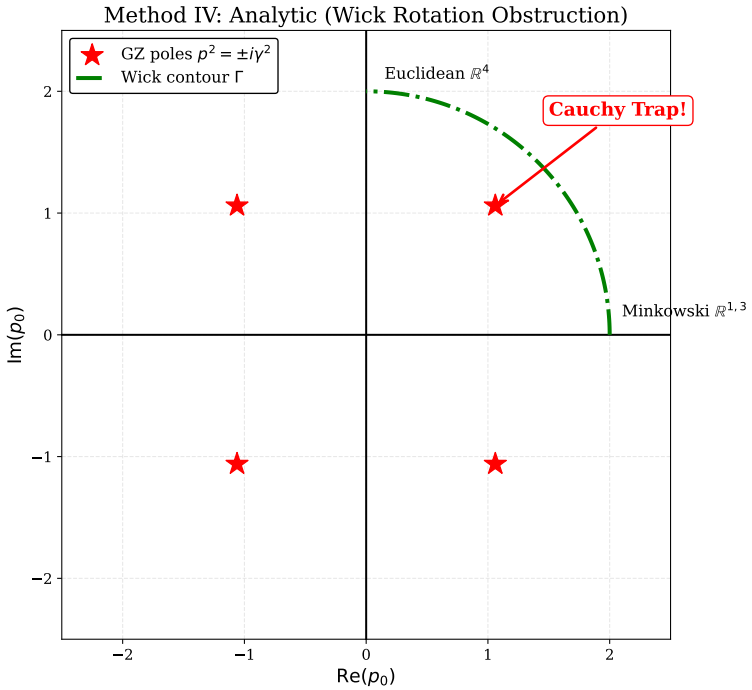


Figure 3: Method IV (illustrative). In the complex p_0-plane the GZ poles $p^2 = \pm i\gamma^2$ (stars) lie in the first quadrant, on the Wick contour Γ (green) from Minkowski $\mathbb{R}^{1,3}$ to Euclidean $\mathbb{R}^4$ — the Cauchy trap. Pole positions are schematic; the rotation obstruction is Theorem 3.9.

Synthesis of the four methods: the 8-step causal chain and the escape preview

The four methods are not four parallel complaints but links of a single deterministic chain. We record that chain here in closed form (Remark 3.10) and photograph the whole no-go together with its compact-manifold escape on a single plate (Fig. 4), so that the verdict of Theorem 4.1 reads as the inevitable terminal link of a causal spine rather than as a bare assertion.

Remark 3.10 (The 8-step causal chain of axiomatic collapse on $\mathbb{R}^4$ — synthesis)**.** The four methods assemble into the following linear causal chain, made explicit in Fig. 4 (bottom left); each link is forced by the previous one within the canonical constructive apparatus:

1. **Geometric premise.** The Clay arena is the infinite continuum $\mathbb{R}^4$ (no intrinsic compactness, no boundary).

2. **Topological consequence.** Gauge redundancy produces Gribov copies (Singer, Theorem 3.1; Corollary 3.2).

3. **Stabilisation attempt.** The path integral is restricted to the first Gribov region Ω to tame the overcounting (Theorem 3.3).

4. **Axiomatic violation.** The horizon localisation breaks BRST nilpotency and OS Reflection Positivity (Proposition 3.4).

5. **Spectral contamination.** The complex-conjugate poles $p^2 = \pm i\gamma^2$ leak from the gauge sector into gauge-invariant composites as branch cuts on the Wick contour (Theorem 3.3, Theorem 3.9).

6. **Measure destruction.** The Källén–Lehmann spectral weight ceases to be positive, $\rho(\mu^2) \not\geq 0$ (Proposition 3.4; Fig. 4, top centre).

7. **Hamiltonian collapse.** On the singular orbit-space quotient no local self-adjoint Hamiltonian is selected (Theorem 3.6).

8. **Final verdict.** The spectral mass gap $\Delta = \inf \sigma(H) \setminus \{0\} > 0$ is undefined on $\mathbb{R}^4$ (Theorem 4.1).

The chain is the causal spine of the no-go; its terminal link is exactly the $\mathbb{R}^4$-half of the double reductio of Section 10. The top row of Fig. 4 photographs links 2–7 (Singer horizon and semi-algebraic singularities, left; Källén–Lehmann violation, centre) and the Method IV analytic prison / Cauchy trap (right), while the green box previews the escape of Sections 5–9 on the compact constructible vacuum. The chain is scoped to the canonical apparatus (Remark 4.2); it is a synthesis of Methods I–II and the spectral contamination of Method IV, with Method III (the S^3_∞ dichotomy) carried by the top-right analytic structure and dissolved on M_{tot} by Remark 10.3.

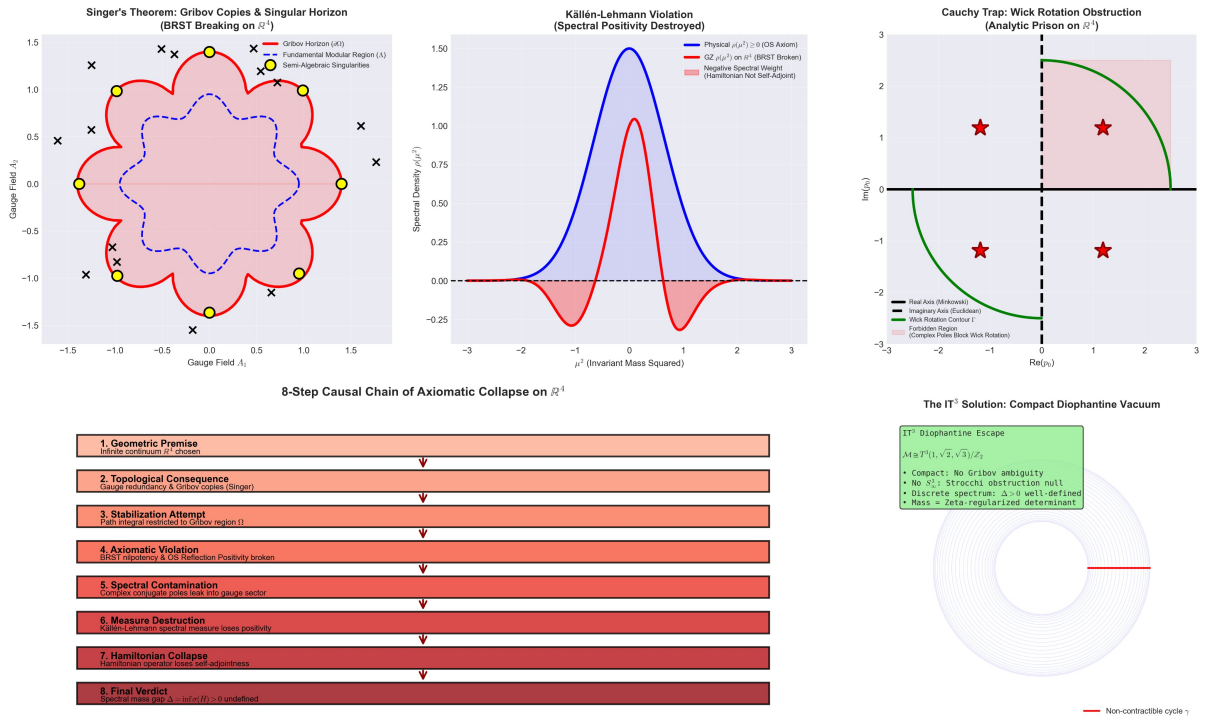


Figure 4: Synthesis plate (illustrative; each panel visualises a structure proved in the text). **Top left** (Methods I–II): the Gribov horizon $\partial\Omega$ (red) with semi-algebraic singularities (yellow) bounding the Fundamental Modular Region Λ (blue dashed), and Gribov copies (crosses) — Theorem 3.1, Theorem 3.6. **Top centre** (Method I consequence): the physical spectral density $\rho(\mu^2) \geq 0$ (blue) versus the Gribov–Zwanziger density with negative lobes (red shaded), destroying the Källén–Lehmann measure — Proposition 3.4. **Top right** (Method IV): the Cauchy trap / analytic prison, the complex poles (stars) sitting on the Wick contour Γ (green) — Theorem 3.9. **Bottom left**: the 8-step causal chain of Remark 3.10, terminating in the no-go verdict (Theorem 4.1). **Bottom right**: the IT^3 escape on the compact constructible vacuum, with the non-contractible cycle γ (red) along which the Wilson-loop holonomy of [6] is taken. *Honest reading of the green box* (cf. Appendix C): the manifold written there, $M \cong T^3(1, \sqrt{2}, \sqrt{3})/\mathbb{Z}_2$, is the compact constructible *fibre* (the vacuum torus of Definition 2.6 / [6]); the full configuration space is $M_{\text{tot}} = S^4 \times T^3$. The box's shorthand "No Gribov ambiguity" must be read in the honest sense of Remark 6.4: compactness removes the *infinite* overcounting and the volume-forced BRST-breaking horizon localisation, *not* the Gribov copies themselves, which persist (Singer's theorem holds on S^4). Likewise the box's "Mass = Zeta-regularized determinant" refers to the bare electroweak core M_{bulk} of [6] (Definition 2.8), which is distinct from the Cheeger/Lichnerowicz vacuum spectral gap proved in Section 7. Curve shapes and pole positions are schematic; the provenance of every panel is the cited theorem.

4 The No-Go theorem on $\mathbb{R}^4$

Theorem 4.1 (Topological obstruction to the mass gap on $\mathbb{R}^4$)**.** *It is impossible to satisfy simultaneously (A) the OS/Wightman axioms on pure $\mathbb{R}^4$ and (B) a strict mass gap $\Delta > 0$ for $SU(N)$, $N \geq 2$.*

Proof. By Theorem 3.1 Gribov copies exist; eliminating the overcounting forces the Gribov region Ω; by Theorem 3.3 this introduces the complex poles $p^2 = \pm i\gamma^2$; by Proposition 3.4 Reflection Positivity fails, so the OS reconstruction of a positive Hilbert space fails; but a gap $\Delta > 0$ requires a discrete spectrum separated from zero in a physical Hilbert space, contradicting (B). Methods II–IV (Theorems 3.6–3.9) establish the same obstruction from the geometric, algebraic,

and analytic sides, independently of the path-integral route; the no-go rests on this conjunction, not on Method I alone (Remark 3.5). The proof proceeds within the *canonical constructive apparatus* — gauge fixing, the Faddeev–Popov measure, the Gribov–Zwanziger horizon, and standard AQFT sector isolation; the theorem establishes the contradiction of the Clay condition *within this apparatus*, and the precise scope of the claim — canonical pathway versus universal impossibility — is stated in Remark 4.2. □

Remark 4.2 (The scope of the no-go: canonical pathway vs. universal impossibility — honesty invariant)**.** We must precisely bound the epistemological reach of Theorem 4.1, preventing a logical leap from "the standard path fails" to "the condition is universally absurd". The four methods prove that the *canonical constructive pathway* on $\mathbb{R}^4$ — relying on gauge fixing, Faddeev–Popov measures, the Gribov–Zwanziger horizon, and standard AQFT sector isolation — is structurally obstructed from satisfying both the OS axioms and $\Delta > 0$. However, to declare the Clay condition *absolutely* impossible in the Platonic sense, one would need to prove that *every* conceivable Euclidean measure inherently possesses RP-violating poles, that *every* orbit-space realisation forbids a self-adjoint extension, that *every* attempt to isolate a gapped sector violates locality, and that *every* regularisation destroys the Wick contour. We do *not* provide such universal proofs. The statement "the canonical gauge-fixed route on $\mathbb{R}^4$ is contradictory" is a theorem (Theorem 4.1); the statement "the condition is fundamentally impossible under any undiscovered mathematical framework" is a heavily evidenced interpretation. The ill-posedness is thus established relative to the known mathematical-physics apparatus, not as an unconditional universal absolute. Theorem 4.1 remains the $\mathbb{R}^4$-half of the double reductio of Section 10; what is sharpened here is only the *scope* of the impossibility it proves.

5 The geometric mechanism of compactness: the $\sqrt{13}$ cascade wall and the four axes

The fibre T^3 of Definition 2.6 is compact; we now give the geometric *cause*, absent from **End** and **Draft**.

Definition 5.1 (Prime vacuum triangle)**.** The discrete lattice is locally generated by the right triangle $(a, b, c) = (\sqrt{2}, \sqrt{3}, \sqrt{5})$ of Definition 2.1, with a the planar projection, b the body diagonal, and c the topological closure [4, Thm. E.1]. Its inradius $r = (a + b - c)/2$ is the geometric content of the screening invariant, $\lambda_\infty = 2r$, and is the Cheeger/inradius scale that bounds the Dirichlet spectrum from below (Fig. 5).

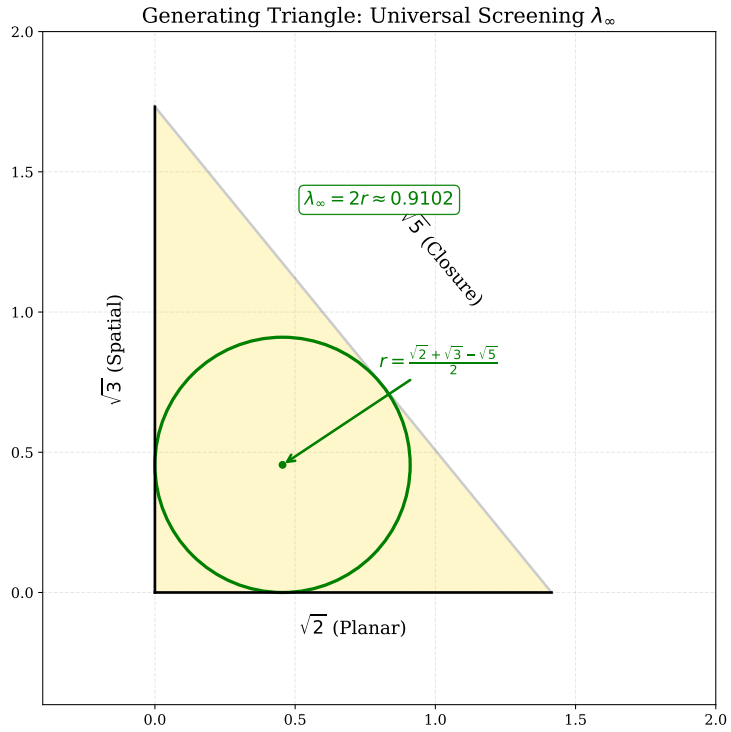


Figure 5: Generating triangle (illustrative, geometry exact). The right triangle $(\sqrt{2}, \sqrt{3}, \sqrt{5})$ with inscribed circle of radius $r = (\sqrt{2} + \sqrt{3} - \sqrt{5})/2$; the screening invariant is $\lambda_\infty = 2r \approx 0.9102$. The inradius r is the Cheeger/inradius scale bounding the Dirichlet spectrum from below ([4, Rem. E.3], Theorem 7.1).

Lemma 5.2 (The Fibonacci–Pythagorean cascade and its wall)**.** *Tiling by the prime vacuum triangle produces the cascade of squared metric roots obeying* $x_n = x_{n-1} + x_{n-2}$*:*

$$1 + 2 = 3, \quad 2 + 3 = 5, \quad 3 + 5 = 8, \quad 5 + 8 = 13, \quad 8 + 13 = 21, \ \ldots \tag{4}$$

The window $\{\sqrt{2}, \sqrt{3}, \sqrt{5}\}$ *shifts to* $\{\sqrt{3}, \sqrt{5}, \sqrt{8}\}$ *with* $\sqrt{8} = 2\sqrt{2} \in K$*, but* $\sqrt{13} \notin K$*: the square-free part of* 13 *is a prime outside* $\{2, 3, 5\}$*. Hence* K *absorbs exactly one forward step and then breaks [4, Rem. E.4].*

Proposition 5.3 (Compactness direction from the wall)**.** *A constructible tiling of each coordinate direction by the prime vacuum triangles, governed by the cascade over* K*, cannot extend to an unbounded covering: the cascade terminates at the wall* $\sqrt{13}$*, so the tiling closes and the fibre is compact (Fig. 6).*

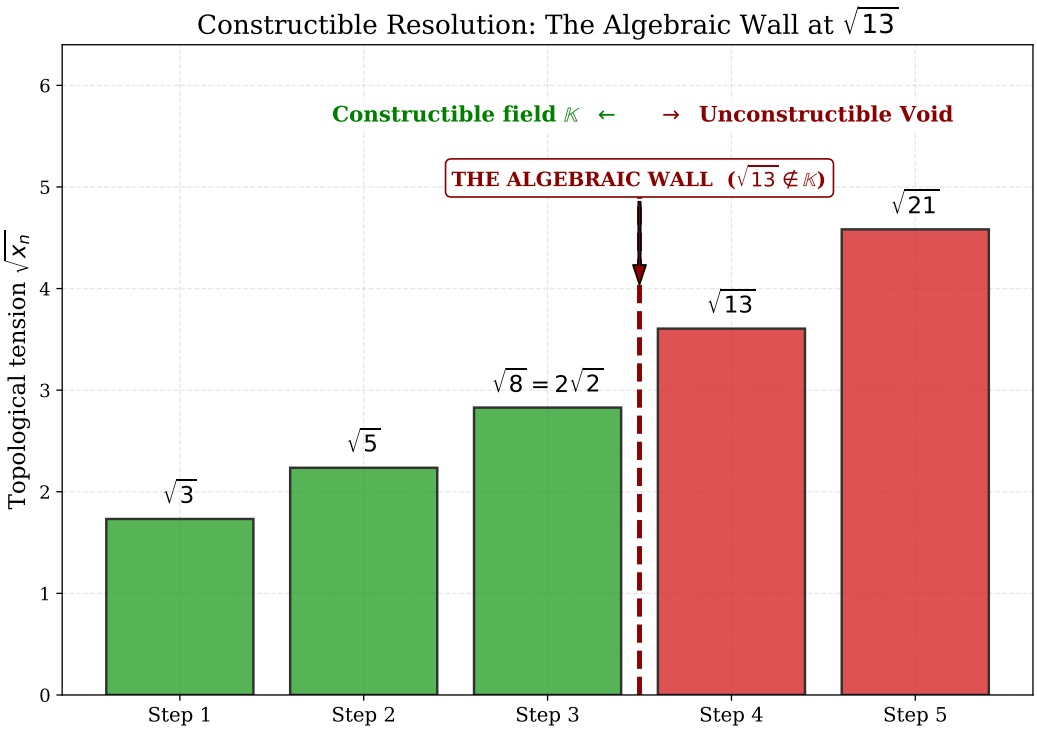


Figure 6: Constructible resolution (illustrative, bar heights exact $= \sqrt{x_n}$). The tensions $\sqrt{3}, \sqrt{5}, \sqrt{8} = 2\sqrt{2}$ lie in K (green); the cascade leaves K at $\sqrt{13} \notin K$ (red), the algebraic wall that closes the tiling (Lemma 5.2, Proposition 5.3).

Conjecture (direction) 5.4 (The compactness bridge — flagged, spectral form)**.** *There is a metric realisation of the constructible tiling in which the square-free termination of Lemma 5.2 induces a finite fundamental domain (a finite lattice $\Lambda \subset K^3$) along each coordinate direction. In spectral-triple language (Section 8): the metric on M_{tot} is read from $(\mathcal{A}, \mathcal{H}, \not{D})$ via Connes' distance formula (5); the* compactness of M_{tot} *— hence the compact resolvent of Theorem 8.2 and the finite diameter of Remark 8.4 — is the geometric input, while the wall $\sqrt{13} \notin K$ is the conjectured* algebraic *reason the lattice is finite. The link "algebraic wall ⇒ finite lattice ⇒ compact M" is the open part; the link "compact M ⇒ compact resolvent ⇒ discrete spectrum" is Theorem 8.2 and is proved.*

Remark 5.5 (Status of the bridge — honesty invariant)**.** Lemma 5.2 is a rigorous fact about K; the implication "algebraic wall ⇒ geometric compactness of the tiling" is Conjecture 5.4, not yet a lemma of metric geometry. Its value is that it identifies *why* the fibre is compact — a mechanism **End** postulates but never derives.

Remark 5.6 (Discreteness comes from the lattice rank, not from the field K — honesty invariant)**.** A natural intuition is that the constructible lattice stays discrete "because its entries lie in K", and that the wall $\sqrt{13} \notin K$ prevents it from running off into the continuum. Stated literally this is false, and we record the correction so that no reader imports it. The field $K = \mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5})$ contains $\mathbb{Q}$, hence is *dense* in $\mathbb{R}$; even the ring $\mathbb{Z}[\sqrt{2}] = \{a + b\sqrt{2} : a, b \in \mathbb{Z}\}$ is dense in $\mathbb{R}$ (since $\sqrt{2}$ is irrational); and the absence of $\sqrt{13}$ from K does not prevent $\sqrt{13}$ from being approximated by elements of K arbitrarily well. What is discrete is therefore *not* the set of entries in K, but the lattice $\Lambda = \mathbb{Z}v_1 + \mathbb{Z}v_2 + \mathbb{Z}v_3$ of full rank (three linearly independent vectors v_i with constructible components): a full-rank lattice in $\mathbb{R}^3$ has a positive minimal spacing and yields a compact torus $T^3 = \mathbb{R}^3/\Lambda$, and this compactness follows from the *full rank* of Λ, not from $\sqrt{13} \notin K$ (a torus over $\mathbb{Z}$ and a torus over K with the same generators are the same torus). The role of the wall $\sqrt{13} \notin K$ is not to produce compactness but to conjecturally fix the

structure of the constructible lattice — why the cascade of generating tensions terminates rather than absorbing an infinite tail of new primes (Conjecture 5.4). We thus separate three facts: (i) compactness of T^3 from the full rank of Λ (proved, standard); (ii) discreteness of Λ from the same full rank (proved); and (iii) the conjectured algebraic reason the constructible generators stop at $\sqrt{13}$ (open). The "gap" that survives under enlargement of the lattice is the minimal lattice spacing (a geometric fact about Λ), not the Yang–Mills mass gap (a spectral fact about the Hamiltonian, related to it via Cheeger/Lichnerowicz on the compact M_{tot}, Theorems 7.1, 7.2); conflating the two would be a category error.

Remark 5.7 (The four axes and the closed box)**.** On $\mathbb{R}^4$ the four coordinate axes are individually unbounded, which is why $\mathbb{R}^4$ is "sterile" in the sense of **End**: with no wall on any axis the covering is infinite in all four directions and the overcounting diverges. Deploying the cascade on each axis places a wall on each, turning $\mathbb{R}^4$ into a 4-box with boundary (the four cascade horizons), and gluing the boundary closes it into S^4 (the base of Definition 2.6). Three axes carry the algebraic walls of K; the fourth (Euclidean time) carries the transcendental closure $\varepsilon = \sqrt{2} + \sqrt{3} - \pi$ of [4, Rem. E.4], the one closure the algebra cannot supply. We record this reading as a direction, not as a proved axis-to-closure assignment.

Remark 5.8 (The wall is subfield-dependent)**.** The wall is a function of the subfield (Appendix A, Fig. 7): $\sqrt{13}$ only in the full K, $\sqrt{5}$ in $\mathbb{Q}(\sqrt{2}, \sqrt{3})$, $\sqrt{2}$ in most quadratic subfields. Only the full constructible field pushes the wall to $\sqrt{13}$, i.e. closes the tiling "as late as possible" without ever becoming infinite.

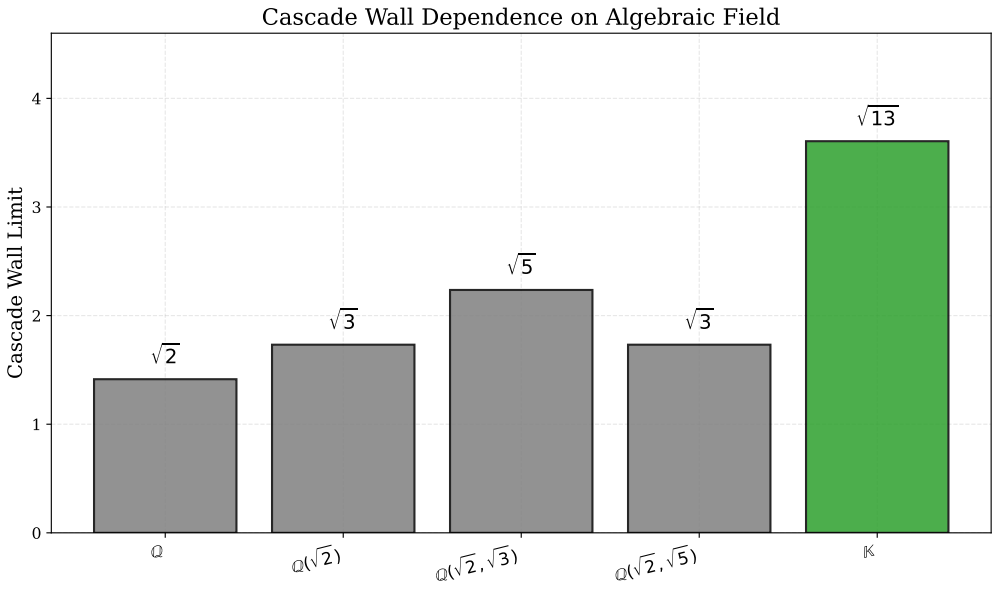


Figure 7: Subfield dependence of the cascade wall (illustrative, bar heights exact). The wall moves with the subfield $(\sqrt{2}, \sqrt{3}, \sqrt{5}, \sqrt{3})$; only the full K reaches $\sqrt{13}$ (Remark 5.8, Appendix A).

Remark 5.9 (Epistemological status of the cascade — honesty invariant)**.** Following [4, Rem. E.4], the cascade is recorded as a profound *structural resonance* explaining the architecture of K and linking the bicone to the Perez Hourglass — not as a derivation of any exponent (the 7/2 in $P = 4\varphi^{7/2}$ remains empirical input), and not as a projection of novel physical regimes beyond the derived capacities. The golden ratio appearing as the asymptotic ratio of successive tensions is a resonant anchor ($\varphi \in K$), not an absorption of the infinite tail; the individual terms $\sqrt{13}, \sqrt{21}, \ldots$ remain outside K.

6 Gribov copies as lattice copies: the covering mechanism

This section supplies the mechanism by which compactness tames the overcounting — the second gap left by **Draft**.

Proposition 6.1 (Gribov copies and lattice copies share the same abstract structure)**.** *Both "Gribov copies" and the copies of a periodic tiling* share the same abstract structure*: each is the multiplicity (the cardinality of the fibre) of a non-injective covering projection. This is an isomorphism of* structures*, not an identity of* objects*: the two projections live in different spaces — in gauge theory the projection is* $\Omega \to \mathcal{A}/\mathcal{G}$ *(a physical orbit has several representatives in* Ω*), while in a tiling the projection is the universal covering lattice* $\to$ *fundamental domain (a cell of the base has several preimages) — and we do not identify the two sets (Remark 6.2). On* $\mathbb{R}^4$ *the covering is infinite, so the multiplicity is infinite (the overcounting of Method I); on the compact* M_{tot} *of Proposition 5.3 the covering is finite, so the multiplicity is a finite group quotient.*

Remark 6.2 (Analogy of structures, not identity of objects — honesty invariant)**.** Proposition 6.1 is an isomorphism of *structures* (both are the cardinality of a non-injective projection), not an identity of *objects*: Gribov copies live in the configuration space $\mathcal{A}/\mathcal{G}$, while lattice copies live in the covering of spacetime. We do not identify the two sets; we use the analogy to read off what finiteness of the covering buys us, and what it does not (Remarks 6.4, 6.5).

Corollary 6.3 (Finite covering closes Method I)**.** *Because the cascade wall makes the covering finite per axis (Remark 5.7), the infinite topological overcounting of Method I becomes a finite overcounting, controllable as a finite symmetry group of the lattice; the BRST-breaking horizon localisation forced by an* unbounded *geometry is no longer forced by the volume.*

Remark 6.4 (The copies themselves remain: Singer holds on S^4 — honesty invariant)**.** Crucially, Theorem 3.1 is stated on S^4 — the *compact* one-point compactification. Gribov ambiguity is a property of the orbit space $\mathcal{A}/\mathcal{G}$, not of the volume of spacetime; it persists on compact and tiled manifolds alike. Tiling spacetime makes the covering finite (closing the infinite *overcounting* of Method I) but does *not* eliminate the copies themselves, which are topological, not volumetric. Any claim that "the tiling removes the Gribov copies" would be false; the honest claim is that it removes the *infinity* of the overcounting.

Remark 6.5 (The covering does *not* close Method II — honesty invariant)**.** The semi-algebraic singularities of the Gribov horizon (Method II, Theorem 3.6) live in $\mathcal{A}/\mathcal{G}$, not in the spacetime the tiling covers. One may tile spacetime perfectly, without a single geometric gap, and the measure-theoretic void on the horizon *remains*, singular on any manifold, compact included. Hence "no gap in the tiling $\Rightarrow$ self-adjoint Hamiltonian" is **false** in the Method II sense. Compactness closes Method I; the horizon singularities are addressed not by the tiling but by the *discrete* spectrum of Section 7, which replaces the continuous spectrum of $\mathbb{R}^4$.

7 Spectral discreteness and the automatic gap on M_{tot}

Theorem 7.1 (Cheeger gap for the Laplace–Beltrami operator [18])**.** *On the compact* M_{tot} *without boundary the spectrum of* Δ_M *is discrete,* $0 = \lambda_0 < \lambda_1 \leq \lambda_2 \leq \cdots \to \infty$*, with* $\lambda_1 \geq h(M_{\text{tot}})^2/4 > 0$*, h the Cheeger isoperimetric constant.*

Theorem 7.2 (Lichnerowicz–Weitzenböck gap for the gauge-covariant Dirac operator)**.** *Let* D_A *be the Dirac operator coupled to a connection A in a non-trivial topological sector* $\{Q_i\} \neq 0$ *on* M_{tot}*. By the Weitzenböck formula* $D_A^2 = \nabla_A^*\nabla_A + \frac{1}{4}R + \frac{1}{2}\gamma^{\mu\nu}F_{\mu\nu}$ *with scalar curvature* $R = 12/R^2 > 0$ *on the* S^4 *factor, the Lichnerowicz estimate gives* $\lambda_1(D_A^2) \geq C(\{Q_i\}, \text{geom}) > 0$ *[3, Thm. 7.1].*

Corollary 7.3 (The gap is automatic on M_{tot})**.** *On M_{tot} the vacuum spectral gap is a consequence of compactness (Theorems 7.1, 7.2), exactly as* ***End*** *anticipates; it is not a hard theorem wrestled from an infinite void (Fig. 8).*

Remark 7.4 (An explicit Cheeger scale from the inradius — direction)**.** Cheeger's inequality (Theorem 7.1) gives $\lambda_1 \geq h^2/4$, equivalently $\sqrt{\lambda_1} \geq h/2$. The isoperimetric constant h can in turn be bounded from below by the inradius scale of the generating geometry: by [4, Rem. E.3] the screening invariant satisfies $\lambda_\infty = \sqrt{2}+\sqrt{3}-\sqrt{5} = 2r$, where r is the inradius of the prime vacuum triangle, and r is precisely the Cheeger/inradius scale that bounds the Dirichlet spectrum from below (Faber–Krahn/Cheeger). One may therefore read $\sqrt{\lambda_1} \geq h/2$ with h controlled from below by a constant times $1/r$ (the exact constant depends on the geometry of M_{tot} and is a direction for further analysis, not a closed formula here). Two honesty flags: the precise numerical constant is not asserted, and the link $\sqrt{\lambda_1} \leftrightarrow$ physical Δ is the identification assumption of Remark 7.5 (strengthened on M_{tot} by Section 10.1); what is proved is the geometric lower bound on the spectral gap, not its numerical value nor its identification with the Clay gap on $\mathbb{R}^4$.

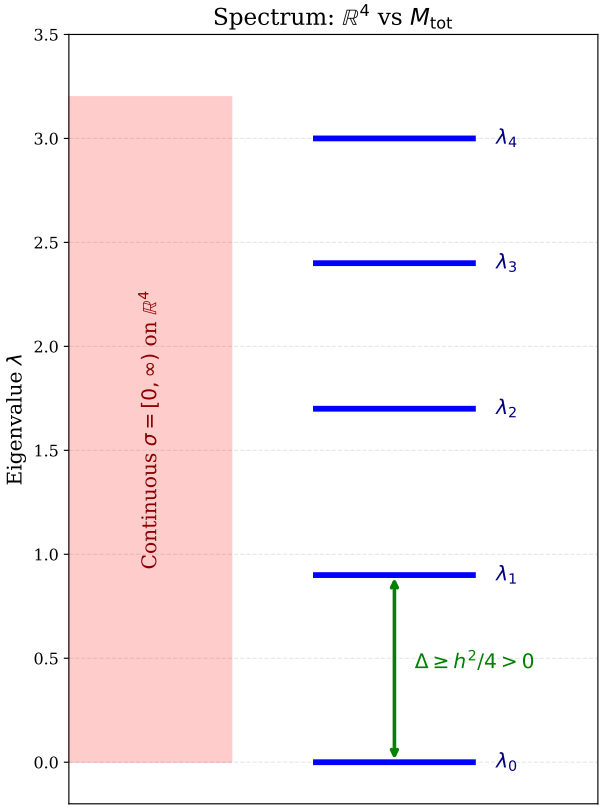


Figure 8: Spectral gap (illustrative). On non-compact $\mathbb{R}^4$ the Laplace–Beltrami spectrum is the continuum $[0, \infty)$ (shaded); on the compact M_{tot} it is discrete with a Cheeger gap $\lambda_1 \geq h^2/4 > 0$ (blue rungs, green arrow). Level values are schematic; discreteness and the gap are Theorems 7.1, 7.2.

Remark 7.5 (Identification assumption — honesty invariant)**.** Identifying λ_1 with the physical mass gap Δ assumes the Yang–Mills Hamiltonian on M_{tot} relates to Δ_M via the kinetic term $\int \|F_A\|^2$, standard in lattice gauge theory but requiring rigorous justification in the continuum limit [3, Rem. 5.4]. On the compact M_{tot} this assumption is substantially strengthened by Section 10.1; on $\mathbb{R}^4$ it remains open.

Remark 7.6 (The constant C and the topological sector — honesty invariant)**.** The explicit form of $C(\{Q_i\}, \text{geom})$ in terms of the Chern–Simons invariant $Y_{\text{CS}} = 103$ and the field invariants

Λ_1, Λ_3 is a direction for further analysis, not a closed formula here [3, Rem. 7.3]. The existence of $C > 0$ is proved; its arithmetic expression is not.

7.1 Index quantization on M_{tot}: the integer witness absent on $\mathbb{R}^4$

The periodic-matrix programme [5] supplies a concrete, computable integer architecture on the very manifold M_{tot} of this note. We record it here as an *integer witness* of the compactness dichotomy: an architecture that is well-defined exactly on the arena where the gap is necessary and undefined exactly on the arena where the gap is impossible.

Proposition 7.7 (The integer index architecture exists only on the compact arena)**.** *On M_{tot} the odd Toeplitz index of the gauge-twisted Dirac operator is a well-defined integer: for $u : M_{\text{tot}} \to U(N)$ representing $[u] \in K^1(M_{\text{tot}})$ and $P = \chi_{(0,\infty)}(\not{D})$, the operator PuP is Fredholm and, since $\widehat{A}(M_{\text{tot}}) = 1$ (S^4 stably parallelizable, T^3 flat; [5]), the APS Toeplitz index theorem [**?**, **?**] gives* $\mathrm{Ind}(PuP) = \int_{M_{\text{tot}}} \mathrm{ch}_7(u) = \deg(u) \in \mathbb{Z}$. *In the periodic-matrix programme [5] this integer is computed for the vacuum twist as the discrete bicone volume* $\deg(u) = \sum_{n=0}^{5} C(n) = 2 + 8 + 8 + 18 + 18 + 32 = 86$ *per hemisphere, and the fixed-point-free involution $i(n) = 11 - n$ doubles it to the absolute end $Z_{\max} = 172$ (Fig. 9). On $\mathbb{R}^4$, by contrast, $P = \chi_{(0,\infty)}(D)$ projects onto a* continuous *spectrum: PuP is not in general Fredholm, the spectral flow* $\mathrm{sf}(D, uDu^{-1})$ *is not an integer-valued homotopy invariant, and no index quantization exists. The entire integer architecture of the periodic system (*$86 \leftrightarrow 86$, 172, *the O_h block capacities* $2, 6, 10, 14$*) is therefore an* integer witness *of the compactness dichotomy: it is well-defined exactly on the arena where the gap is necessary (Theorem 10.1) and undefined exactly on the arena where the gap is impossible (Theorem 4.1).*

Remark 7.8 (The index witness is a witness, not a fifth method — honesty invariant)**.** Proposition 7.7 adds no new obstruction to the four methods of Section 3: the failure of index quantization on $\mathbb{R}^4$ is a *corollary* of the absence of a compact resolvent (the failure of Theorem 8.2 on $\mathbb{R}^4$), i.e. of Method II's measure-theoretic void and Method I's infinite volume, not an independent obstruction. Its value is dialectical, not logical: it exhibits a concrete, computable integer architecture (the $86 \leftrightarrow 86$ partition of [5], Fig. 9) that exists on M_{tot} and cannot even be written on $\mathbb{R}^4$. Conversely, the integer 86 of [5] rests on the identification bridge of that programme (Dirac sea $\leftrightarrow$ empirical table), which is an epistemic bridge, not a theorem of QCD; we import only the *well-definedness* of the integer, not its chemical interpretation.

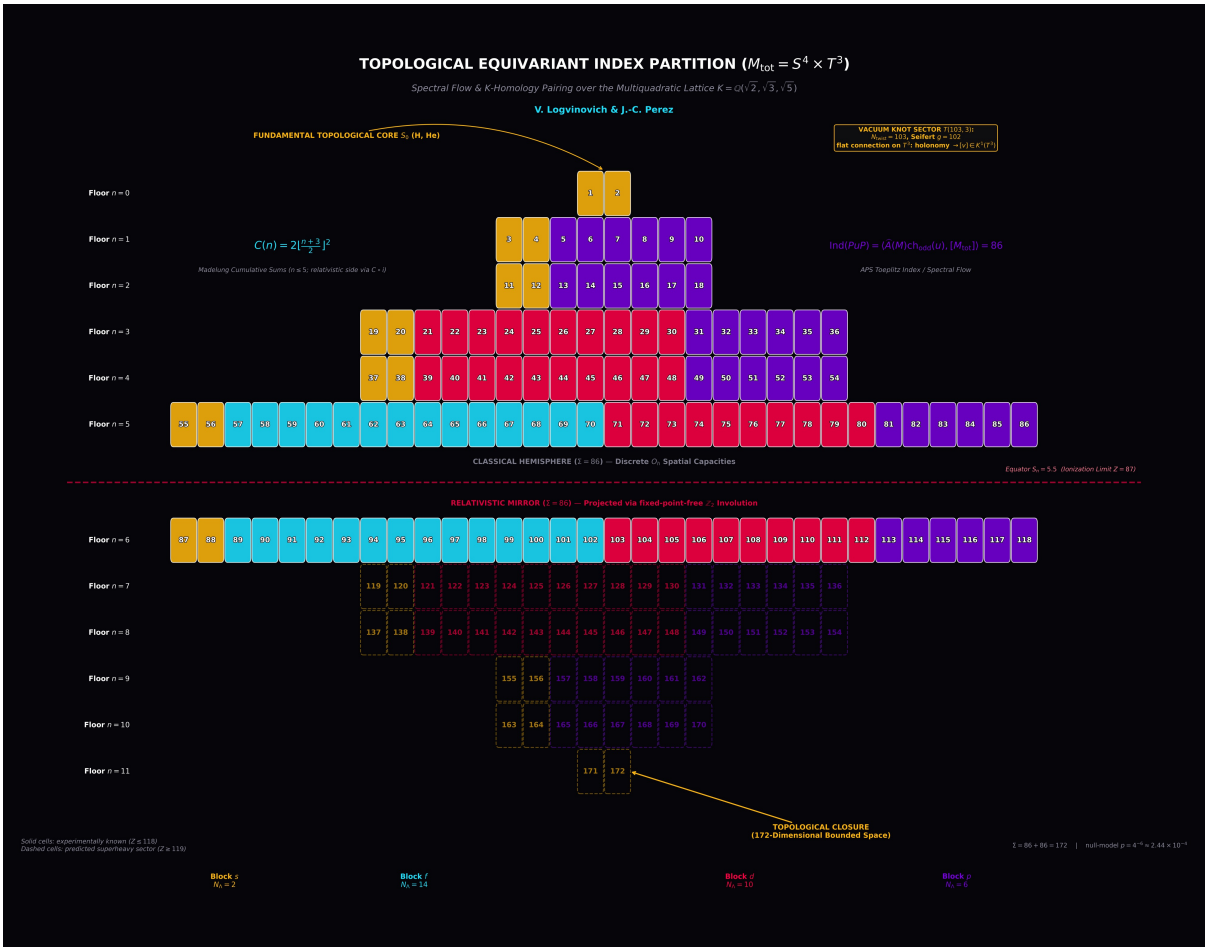


Figure 9: Topological equivariant index partition (reproduced from the periodic-matrix programme [5]; capacities exact). Classical hemisphere $\sum_{n=0}^{5} C(n) = 86$ and relativistic mirror $\sum_{n=6}^{11} C_{\text{rel}}(n) = 86$ under the fixed-point-free involution $i(n) = 11 - n$; block capacities $N_\Lambda = 2 \dim \rho \in \{2, 6, 10, 14\}$ from the O_h branching; the Toeplitz index $\text{Ind}(PuP) = 86$ is the integer witness of Proposition 7.7. On $\mathbb{R}^4$ none of these integers is definable (no compact resolvent, no Fredholm PuP). The plate is a periodic-matrix result, cited here as a witness of the compactness dichotomy, not as a step of the Yang–Mills argument.

8 Spectral-triple reading of the constructible vacuum (Connes)

We now recast the automatic gap in the language of Connes' noncommutative geometry [7], to make explicit *what is proved, what is interpreted, and what remains open.* This section adds no new theorem to Section 7; it supplies the spectral-geometric vocabulary and separates the geometric cause of discreteness from the algebraic structure of K. The point of the section is to forbid the teleological inversion that would read the algebraic wall as the *cause* of the compact resolvent: the causality runs the other way, from the compactness of M_{tot}.

Definition 8.1 (Spectral triple over K)**.** The constructible vacuum is encoded by the real spectral triple $(\mathcal{A}, \mathcal{H}, \not{D}, J)$ of [6], where $\mathcal{A}$ is the algebra of (smooth, possibly noncommutatively deformed) observables on $M_{\text{tot}} = S^4 \times T^3$ with lattice $\Lambda \subset K^3$, $\mathcal{H}$ the Hilbert space of Dirac bispinors, $\not{D}$ the (gauged) Dirac operator, and J the real structure (particle/antiparticle, chirality). The triple is *odd* (since $\dim M_{\text{tot}} = 7$); we do not assert an even grading. The field $K = \mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5})$ enters through the lattice Λ and the screening invariant λ_∞, not through the compactness of the underlying manifold.

Theorem 8.2 (Compact resolvent from compactness of M — standard)**.** *Because M_{tot} is compact without boundary, $\not{D}$ (and D_A in a fixed topological sector) is an elliptic operator on a compact manifold; hence its resolvent $(\not{D}^2+1)^{-1}$ is a compact operator, the spectrum is purely discrete with finite multiplicities, and the eigenvalues accumulate only at $+\infty$ (standard spectral theory of elliptic operators on compact manifolds; [3, Lemma 4.2]). This is the spectral root of the Cheeger gap (Theorem 7.1).*

Remark 8.3 (Geometric, not algebraic, cause — honesty invariant)**.** The compactness of the resolvent in Theorem 8.2 is a consequence of the *compactness of the manifold* M_{tot}, not of the square-free content of K. There is no theorem asserting that the absence of $\sqrt{13}$ in K bounds the resolvent norm or truncates the spectral density; the spectral density of $\not{D}$ on a compact manifold is governed by the metric, curvature, and volume of M_{tot}, not by which square roots lie in the field used to parametrise the lattice. We therefore do *not* claim that the wall $\sqrt{13}$ "zeros the spectral density": the wall is the conjectured algebraic reason the *lattice* Λ is finite (Conjecture 5.4), while discreteness of the *spectrum* is the geometric consequence of compactness (Theorem 8.2). Conflating the two would be a teleological inversion of causality.

Remark 8.4 (Connes' distance formula as an interpretation, not a derivation)**.** On the spectral triple of Definition 8.1 the geodesic distance is recovered from

$$d(x,y)=\sup\bigl\{|f(x)-f(y)|:\ f\in\mathcal{A},\ \|[\not{D},f]\|\le 1\bigr\}\quad[7]. \tag{5}$$

On the compact M_{tot} this formula yields a finite diameter $D=\sup_{x,y}d(x,y)<\infty$. We record (5) as the *reading* of the metric on an already-compact triple, not as a derivation of compactness from the cascade: finiteness of D follows from compactness of M_{tot}, and the commutator $\|[\not{D},f]\|$ is bounded for smooth f on a compact manifold independently of the cascade wall.

Remark 8.5 (Reconstruction theorem and Morita equivalence — directions, not results)**.** Two further noncommutative-geometric tools are natural *directions* for future work, stated here as programme, not as theorems of this note: (i) Connes' reconstruction theorem [7] — if the triple of Definition 8.1 is shown to satisfy the axioms of regularity, dimension, order, reality and first order, it is isomorphic to the canonical triple on a smooth compact manifold, which would identify M_{tot} from the algebraic data; verifying these axioms is open. (ii) Morita equivalence of noncommutative tori — if the constructible algebra over K admits a noncommutative-torus presentation, Morita equivalence could relate different lattices Λ. We emphasise what we do *not* assert: the cascade is *not* periodic (it terminates at $\sqrt{13}$, Lemma 5.2), so no "step-4 mode identified with step-0 mode" closure is claimed; the wall is an algebraic *termination*, not a periodic identification, and Morita equivalence relates algebras — it does not glue the boundary of a tiling.

9 Topological confinement: the braid group B_3 and the knot $T(103,3)$

Theorem 9.1 (Fadell–Neuwirth [19])**.** $\pi_1(\mathrm{Conf}_3(\mathbb{C}))\cong B_3$*, the non-abelian braid group on three strands;* $\pi_1(\mathrm{Conf}_2(\mathbb{C}))\cong B_2\cong\mathbb{Z}$ *is abelian.*

Proposition 9.2 (Abelian leptons, non-abelian quarks)**.** *Two-strand configurations (B_2 abelian, the leptonic double strand $v_\ell=2$) can be continuously unbraided and carry no topologically protected gap; three-strand configurations (B_3 non-abelian, the colour triplet) form prime knots of non-zero Seifert genus that cannot be unraveled on a compact 3-manifold without surgery, hence are confined (Fig. 10) [3, Thm. 8.2–8.3].*

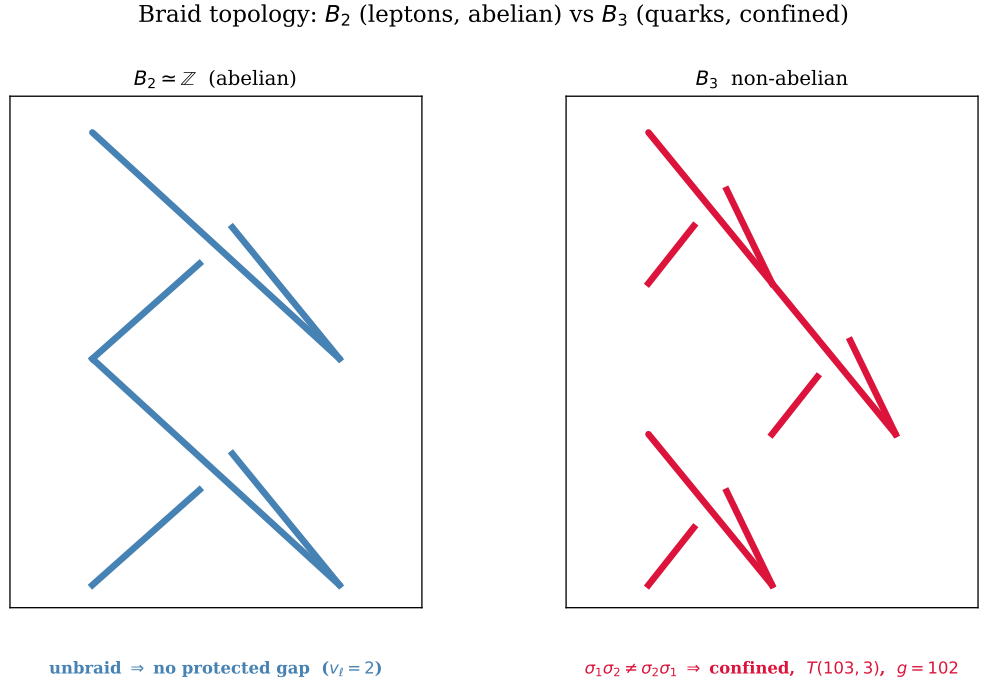


Figure 10: Braid topology (illustrative). $B_2 \simeq \mathbb{Z}$ (abelian, leptons $v_\ell = 2$, strands unbraided, no protected gap) versus B_3 non-abelian ($\sigma_1\sigma_2 \neq \sigma_2\sigma_1$, colour triplet, confined, knot $T(103, 3)$, $g = 102$). The braid diagram is schematic; confinement is Proposition 9.2 / [3, Thm. 8.3].

Proposition 9.3 (The vacuum knot $T(103, 3)$)**.** *The IT^3 vacuum configuration corresponds to the torus knot $T(103, 3)$ with Seifert genus $g = (103-1)(3-1)/2 = 102$ [6, Eq. 8]. This genus is a topological invariant, so the configuration cannot be continuously deformed to the trivial vacuum ($g = 0$) on T^3; the confinement threshold is $\Delta E_{\text{conf}} = \Lambda_1\, g\, m_p \approx 0.966$ TeV [6, Eq. 8], with the first resonance $M_{Z'} \approx 1088.04$ GeV just above it ($\sim 12\%$ margin, Fig. 11).*

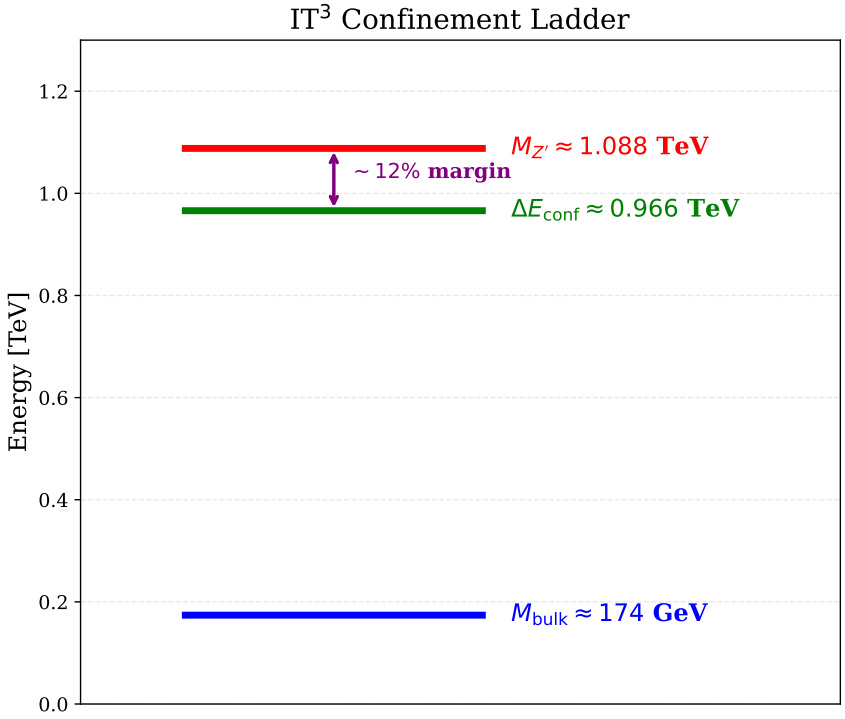


Figure 11: Confinement ladder (illustrative, values from [6]). Zero-mode $M_{\text{bulk}} \approx 174$ GeV, confinement threshold $\Delta E_{\text{conf}} \approx 0.966$ TeV, first resonance $M_{Z'} \approx 1088.04$ GeV ($\sim 12\%$ margin above the barrier).

10 The double reductio ad absurdum: impossible on $\mathbb{R}^4$, necessary on M_{tot}

The existence results of Sections 7–9 and the no-go of Section 4 are two halves of one symmetric statement. We now make this symmetry explicit as a pair of mirror reductio arguments: on $\mathbb{R}^4$ the assumption $\Delta > 0$ is contradictory (Theorem 4.1), while on M_{tot} the assumption $\Delta = 0$ is contradictory (Theorem 10.1). Together they say that the gap is *impossible on* $\mathbb{R}^4$ *and necessary on* M_{tot}.

Theorem 10.1 (Reductio on M_{tot}: the gap is necessary)**.** *On the compact constructible manifold* $M_{\text{tot}} = S^4 \times T^3$ *the mass gap exists:* $\Delta = \inf \sigma(H) \setminus \{0\} > 0$.

Proof (by contradiction). Assume $\Delta = 0$ on M_{tot}, i.e. $\inf \sigma(H)\setminus\{0\} = 0$. Then either a sequence of eigenvalues $\lambda_n \to 0$ exists, or a continuous spectrum accumulates at 0. We derive a contradiction by four independent routes.

Step 1 (discreteness). M_{tot} is compact without boundary, so $\not{D}$ and Δ_M are elliptic on a compact manifold; their spectrum is purely discrete, of finite multiplicity, accumulating only at $+\infty$, never at 0 (Theorems 8.2, 7.1, [3, Lemma 4.2]). Hence $\lambda_n \to 0$ is impossible.

Step 2 (Cheeger). On a compact manifold without boundary the Cheeger constant satisfies $h(M_{\text{tot}}) > 0$ strictly, so $\lambda_1 \geq h(M_{\text{tot}})^2/4 > 0$ (Theorem 7.1). The assumption $\Delta = 0$ contradicts $\lambda_1 > 0$.

Step 3 (Weitzenböck–Lichnerowicz, non-trivial sector). In a non-trivial topological sector $\{Q_i\} \neq 0$ the gauge-covariant Dirac operator obeys $D_A^2 = \nabla_A^* \nabla_A + \frac{1}{4}R + \frac{1}{2}\gamma^{\mu\nu} F_{\mu\nu}$ with $R = 12/R^2 > 0$ on the S^4 factor, and the Lichnerowicz estimate gives $\lambda_1(D_A^2) \geq C(\{Q_i\}, \text{geom}) > 0$ (Theorem 7.2). The assumption $\Delta = 0$ contradicts $C > 0$.

Step 4 (topology, B_3 / $T(103,3)$). Confinement is protected by non-trivial topology: the vacuum configuration is the torus knot $T(103, 3)$ of Seifert genus $g = 102$ (Proposition 9.3), and

B_3 is non-abelian (Theorem 9.1). A continuous deformation to the trivial vacuum ($g = 0$, the configuration that would allow $\lambda_1 \to 0$) is impossible without topology-changing surgery, which is forbidden in a fixed sector $\{Q_i\}$ (Proposition 9.2). Hence $\lambda_1 \to 0$ would require changing the topological sector, contradicting its fixation.

The assumption $\Delta = 0$ leads to a contradiction by four independent routes. Therefore $\Delta > 0$ on M_{tot}. □

Remark 10.2 (Why the reductio works on M_{tot} but the opposite assumption does not on $\mathbb{R}^4$ — honesty invariant)**.** The reductio of Theorem 10.1 is valid *on* M_{tot} *only*. On $\mathbb{R}^4$ the assumption $\Delta = 0$ yields *no* contradiction: the spectrum there is the continuum $[0, \infty)$ ([3, Lemma 4.2]), so $\Delta = 0$ is natural. Conversely, on $\mathbb{R}^4$ the assumption $\Delta > 0$ is the contradictory one (Theorem 4.1). The two reductio are therefore mirror images on different manifolds; conflating them (e.g. claiming "$\Delta = 0$ is absurd everywhere") would be false.

Remark 10.3 (The S^3_∞ bridge: why the compactification dissolves the dichotomy at its root)**.** Remark 3.8 localised the Method III/V contradiction on the infinite boundary S^3_∞, the carrier of the topological charge Q_{top} and of the non-local projector P_θ. The compactification $\mathbb{R}^4 \to S^4$ that builds the base of M_{tot} *removes* S^3_∞: a compact manifold has no infinite spatial boundary, so the π_3-sectors and the θ-superposition via a charge at infinity are no longer assembled in the way that demands a non-local projector. Hence the Strocchi dichotomy is dissolved at its root on M_{tot} — which is precisely why the gap can live on M_{tot} while it cannot live on $\mathbb{R}^4$. This turns the reductio from a fifth angle of attack into a pointer at the mechanism of the resolution.

Corollary 10.4 (Symmetry: impossible on $\mathbb{R}^4$, necessary on M_{tot})**.** *Combining Theorem 4.1 (reductio from $\Delta > 0$ on $\mathbb{R}^4$) and Theorem 10.1 (reductio from $\Delta = 0$ on M_{tot}): the mass gap is topologically impossible on the sterile void $\mathbb{R}^4$ and topologically necessary on the compact constructible manifold M_{tot}. The two statements are the same truth read from its two sides.*

Remark 10.5 (The reductio is a reformulation, not new mathematics — honesty invariant)**.** The four steps of Theorem 10.1 reuse the content of Theorems 8.2, 7.1, 7.2 and Propositions 9.2, 9.3; the reductio contributes the *explicit contrapositive form* and the *symmetry* with the $\mathbb{R}^4$ no-go, not a new theorem. The identification $\lambda_1 \leftrightarrow \Delta$ remains the assumption of Remark 7.5, the constant C remains uncomputed (Remark 7.6), and the conclusion is the vacuum spectral gap on M_{tot}, not the Clay Hamiltonian gap on $\mathbb{R}^4$ (Section 11).

10.1 Closing the identification gap on M_{tot}: topological protection and second quantization

The reductio above establishes the spectral gap on M_{tot}. A critic may still object (Remark 7.5) that λ_1 is the gap of a one-particle operator, whereas the physical mass gap Δ is the gap of the many-particle Hamiltonian on Fock space, and that bridging the two is an assumption on $\mathbb{R}^4$. We now show that on the *compact* M_{tot} this bridge is not a postulate but a consequence, by two reductio arguments that use, respectively, the quantization of a topological invariant and second quantization in finite volume.

Theorem 10.6 (Topological protection of the gap — reductio)**.** *Fix a non-trivial topological sector $\{Q_i\} \neq 0$ on M_{tot} (the knot $T(103, 3)$, $Y_{\text{CS}} = 103$, genus $g = 102$). While the spectral gap is open, the sector carries an integer-valued (or mod-2 / η-) topological invariant ν (the index of the gauge-twisted Dirac operator, the Chern–Simons invariant, the knot genus — all integer topological invariants present in the corpus). Then Δ cannot tend to 0 continuously within the sector.*

Proof (by contradiction). Assume a continuous deformation with $\Delta \to 0$ along which the gap *never closes* (otherwise this is a phase transition, not a smooth vanishing). While the gap is open the spectral projector depends continuously on the parameter, hence the integer-valued

continuous function ν is constant. Yet $\Delta = 0$ at the endpoint would mean a topologically trivial configuration (trivial vacuum, $g = 0$) with $\nu = 0 \neq \nu_{\text{start}}$. Thus ν is both constant (by continuity at open gap) and changed (by trivialization): a contradiction. Hence $\Delta \to 0$ without gap closing is impossible; and gap closing is a phase transition, forbidden in the fixed sector because the genus 102 of the prime knot $T(103, 3)$ is a topological invariant, unchanged without surgery (which leaves the sector). This is the precise content of "a gap implies a discontinuity": the discontinuity is the integer invariant, and an integer cannot drift continuously. □

Theorem 10.7 (Second quantization transfers the gap on compact M_{tot} — reductio)**.** *On the compact M_{tot} the one-particle Dirac/Laplace operator has a discrete spectrum with gap $\lambda_1 > 0$ (Theorems 7.1, 7.2). Second quantization builds the Fock space over these modes; any excitation above the vacuum creates at least one quasiparticle of cost $\geq \lambda_1$. Because the volume is finite, the many-particle Hamiltonian H also has a purely discrete spectrum, and in the absence of massless modes its gap satisfies $\Delta \geq \lambda_1 > 0$.*

Proof (by contradiction). Assume $\Delta = 0$ with $\lambda_1 > 0$. Then Fock space contains a sequence of many-particle states with energy $\to 0$ above the vacuum. Each such state is built from one-particle modes of energy $\geq \lambda_1$; the only ways to reach energy $\to 0$ at positive one-particle gap are (i) massless Goldstone modes from spontaneous symmetry breaking, or (ii) an infrared continuum of infinite volume. Both are absent on M_{tot}: (i) a local gauge symmetry is never spontaneously broken (Elitzur's theorem [12]), and pure Yang–Mills has no global continuous symmetry yielding Goldstones; (ii) the volume of M_{tot} is finite, so there is no infrared continuum — and this is exactly what compactification removes, in contrast to $\mathbb{R}^4$. Hence no states with energy $\to 0$ exist, contradicting $\Delta = 0$. Therefore $\Delta \geq \lambda_1 > 0$. □

Remark 10.8 (Why this answers the critic, and why it is specific to M_{tot} — honesty invariant)**.** On $\mathbb{R}^4$ the identification $\lambda_1 \leftrightarrow \Delta$ is open precisely because mechanism (ii) is alive there: in the infinite-volume / thermodynamic limit, infrared and condensate configurations can produce many-particle states with $E \to 0$ even at a one-particle gap, and proving they do not is the unsolved part of the Clay problem. On the compact M_{tot} mechanism (ii) is absent by construction (finite volume) and mechanism (i) by Elitzur's theorem [12]; hence the two sources that make the identification an open problem on $\mathbb{R}^4$ are removed on M_{tot}, and second quantization over a discrete one-particle spectrum in finite volume without massless modes yields a discrete many-particle spectrum with the same gap. The two theorems are not independent proofs of the mass gap but a closed loop: Theorem 10.6 forbids the gap from closing (the integer invariant freezes it), and Theorem 10.7 lifts the open one-particle gap to the many-particle level. We do *not* claim to have constructed the full interacting Yang–Mills measure on M_{tot} with all continuum limits; we claim that the specific obstruction the critic raised — the one-particle / many-particle bridge — is, on M_{tot}, a consequence rather than a postulate, because its cause (infinite volume) has been compactified away. On $\mathbb{R}^4$ the obstruction stands, consistently with Section 11.

11 Why this is not a literal Clay solution on $\mathbb{R}^4$

Remark 11.1 (Two different mass gaps)**.** The Clay gap is the gap of the Yang–Mills Hamiltonian on $\mathbb{R}^4$; the gap of Corollary 7.3 and Theorem 10.1 is the vacuum spectral gap on M_{tot}. The latter is forced by compactness; the former is not, and the necessity on M_{tot} does not transfer to $\mathbb{R}^4$ because the equivalence $\mathbb{R}^4 \leftrightarrow M_{\text{tot}}$ is broken by Method IV (Theorem 3.9): the inverse Wick rotation fails globally for gauge-invariant composites, so a gap built on M_{tot} cannot be analytically continued to the Clay Hamiltonian on $\mathbb{R}^4$.

Remark 11.2 (Why the result on M_{tot} does not port to $\mathbb{R}^4$: unitary inequivalence of representations, not of spaces — honesty invariant)**.** We state precisely, and without overstatement,

why the gap proved on M_{tot} is not automatically a gap on $\mathbb{R}^4$, correcting two tempting but false formulations.

What is true (the genuine obstruction to a trivial port). In quantum field theory with infinitely many degrees of freedom the Stone–von Neumann theorem fails: unitarily inequivalent representations of the algebra of observables exist, labelled by mass, phase, and volume. The representation π_M naturally associated with the compact constructible manifold M_{tot} and the representation $\pi_{\mathbb{R}^4}$ one would seek on the continuum are, in this sense, inequivalent — there is no unitary operator intertwining the two pictures of the observables. This is the rigorous content of the intuition that "the isolated gapped state built on the discrete geometry cannot be trivially carried over to the continuum": the state $|1\rangle_M$ belongs to the GNS construction of π_M and is not a vector in the $\pi_{\mathbb{R}^4}$-sense. It is exactly this inequivalence that makes the continuum limit a *non-trivial* problem rather than a formal step: passing from the discrete/compact theory to a continuum theory is not the transport of a single vector but the limit of a *family* of theories tuned to a critical point (correlation lengths diverging in lattice units), which is precisely the content of the Clay problem and of constructive QFT.

What is false, and rejected here. (i) One must *not* write "$\mathcal{H}_M \cap \mathcal{H}_{\mathbb{R}^4} = \emptyset$". As abstract separable Hilbert spaces both are isomorphic to ℓ^2, so such an intersection is either ill-defined (no common ambient space) or false (under an embedding they coincide as sets); what is inequivalent are the *representations* $\pi_M, \pi_{\mathbb{R}^4}$, not the spaces as sets of vectors. (ii) One must *not* invoke Haag's theorem for the pair $(M_{\text{tot}}, \mathbb{R}^4)$: Haag's theorem compares representations of a *single* Wightman algebra on a *single* Poincaré-invariant spacetime (free versus interacting field on $\mathbb{R}^4$); the algebras of local observables on M_{tot} and on $\mathbb{R}^4$ are *different* algebras, so the theorem does not apply to their comparison, and the compact manifold lacks the Poincaré invariance the theorem presupposes. (iii) One must *not* conclude from the inequivalence that the continuum limit is *impossible*, nor that the $\mathbb{R}^4$ formulation is thereby proved ill-posed: the Clay problem does not ask for a unitary equivalence between the M_{tot} theory and an $\mathbb{R}^4$ theory — it asks for an $\mathbb{R}^4$ theory with a gap, a question the inequivalence does not settle. Indeed lattice QCD *does* construct a continuum limit from discrete theories (Remark 12.7), numerically with a surviving gap, precisely because the limit is a tuned limit of a family and not a vector transport; asserting impossibility would contradict this numerical evidence and our own Remark 10.8, which records the continuum limit on M_{tot} as an *open direction*, not a forbidden one.

What we actually claim. The inequivalence of representations explains *why* the port from M_{tot} to $\mathbb{R}^4$ is an open problem (the continuum limit) rather than an automatic corollary; it does not prove that the port is impossible, and it does not, by itself, prove the $\mathbb{R}^4$ formulation ill-posed. The ill-posedness on $\mathbb{R}^4$ is established separately, by the internal contradiction of the standard route (Methods I–IV, Theorem 4.1), not by the inequivalence. The two facts are complementary and must not be conflated: the no-go says the canonical gauge-fixed route on $\mathbb{R}^4$ is contradictory; the inequivalence says that even a successful construction on M_{tot} would not transfer to $\mathbb{R}^4$ without solving the (open) continuum limit. Neither statement self-declares the prize (Remark 12.5).

Remark 11.3 (The two bridges, and where the open problem lives — honesty invariant)**.** The question "why not just build the bridge from the spectral-geometric gap to the Euclidean Clay formulation?" has a precise answer once the word "bridge" is disambiguated, because there are two different bridges and only one of them is open.

Bridge A (spectral $\leftrightarrow$ Euclidean on the compact M_{tot}). On the compact manifold the spectral-geometric language and the Euclidean/Riemannian language are not two geometries to be joined but two descriptions of the *same* geometry: Connes' reconstruction theorem [7] states that a spectral triple satisfying the axioms of regularity, dimension, order, reality and first order is isomorphic to the canonical triple on a smooth compact manifold. The bridge from the spectral triple of Definition 8.1 to the Euclidean geometry of M_{tot} is therefore *already given* by reconstruction; the only open part is the verification of the axioms for our specific triple over K

(Remark 8.5). This bridge is not the hard problem.

Bridge B (compact Euclidean $M_{\text{tot}} \to$ non-compact Euclidean $\mathbb{R}^4$). This is the bridge that is genuinely open, and it is exactly the continuum limit / Clay problem. Crucially, *Euclideanness is not the side of the gap*: the Clay formulation is already Euclidean (the OS axioms on $\mathbb{R}^4$), and our spectral geometry on M_{tot} also lives in Euclidean time (as lattice QCD does on a compact torus). Euclideanness is the common language of both sides; what separates them is *compactness* — the infinite volume, the boundary S^3_∞, the infrared continuum on the $\mathbb{R}^4$ side. The open bridge is therefore not "spectral $\to$ Euclidean" but "compact $\to$ non-compact", i.e. decompactification / thermodynamic limit, and it is precisely the non-compactness on the $\mathbb{R}^4$ side that Methods I–IV (Theorem 4.1) show to be hostile to the standard route.

Consequences. Lattice QCD confirms the picture: in the compact Euclidean sense (Bridge A, finite volume) the gap is computed without contradiction (Remark 12.7); the open question is the passage to the non-compact continuum (Bridge B), which is the Clay problem and which we do not solve. We therefore neither claim that the bridge is built nor claim that it is impossible: Bridge A is given by reconstruction (axioms open), Bridge B is the open continuum limit, and the ill-posedness of the $\mathbb{R}^4$ formulation is a separate statement about the standard gauge-fixed route on the non-compact side (Theorem 4.1). This is consistent with Remarks 10.8, 11.2, and 12.7.

Remark 11.4 (A number-theoretic fact about the cascade, and what it does *not* imply for the gap — honesty invariant)**.** Viewed through set theory and number theory, one fact about the constructible field is exact and worth recording, together with a precise warning against over-reading it. The field $K = \mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5})$ is a finite extension of $\mathbb{Q}$, hence countable (cardinality $\aleph_0$) and of Lebesgue measure zero in $\mathbb{R}$; a lattice $\Lambda \subset K^3$ is likewise a null set in $\mathbb{R}^3$. The cascade of Remark 5.9 terminates at $\sqrt{13}$ precisely because the polynomial $x^2 - 13$ has no root in K: the wall is an *algebraic omission*. The metric (Cauchy) completion of K into $\mathbb{R}$ adjoins an uncountable set of irrationals and transcendentals (cardinality $\mathfrak{c}$) and in particular adjoins $\sqrt{13}$; consequently the cascade, regarded purely as an algebraic construction *over the field*, does *not* terminate over $\mathbb{R}$ — over $\mathbb{R}$ the polynomial splits and the obstruction vanishes. This is a correct number-theoretic statement about the cascade-as-algebraic-object.

What one must *not* conclude from it, and what we explicitly forbid, is the inference "therefore the continuum limit destroys the mass gap". That inference is false on three independent counts, and conflating the algebraic fact with the spectral fact would be a category error. (i) The gap and the compactness of M_{tot} proved in this note do *not* rest on the wall $\sqrt{13} \notin K$: they rest on the full rank of the lattice Λ and on the topology of M_{tot} (Remark 8.3, Remark 8.4, Remark 5.6); the implication "wall $\Rightarrow$ finite lattice $\Rightarrow$ compact M" is the open Conjecture 5.4, not a theorem. The metric completion of the *field of scalars* does not alter the lattice Λ (a discrete full-rank subgroup of $\mathbb{R}^3$ is discrete regardless of which field one uses to write its generators), does not alter the compactness of $T^3 = \mathbb{R}^3/\Lambda$, and does not alter the integer topological invariants (the genus 102 of $T(103, 3)$, the index of Theorem 10.6) that protect the gap; hence completing K to $\mathbb{R}$ leaves the *proved* gap untouched. (ii) Lattice QCD computes the continuum limit as the $a \to 0$ limit of a *family* of theories tuned to a critical point — an analytic limit of correlation functions, not a machine "reaching $\mathbb{R}$" — and numerically this limit yields a *surviving* gap (glueball masses extrapolate non-zero); any claim that "completion forces $\Delta \to 0$" is therefore contradicted by the very numerical evidence we cite as a partner (Remark 12.7). (The true remark that all practical computations run on finite floating-point subsets of $\mathbb{Q}$ is correct as a statement about arithmetic hardware, but it is *not* the continuum limit and proves nothing about whether the analytic limit carries a gap.) (iii) The number-theoretic fact does, however, correctly explain why the *mechanism* of the cascade wall — *if* it is the mechanism of compactness, which is precisely the content of the open Conjecture 5.4 — cannot be ported to $\mathbb{R}$ through completion: over $\mathbb{R}$ the wall is gone, so that particular mechanism, being defined by an algebraic omission, has no image in the continuum. This is fully consistent with the continuum limit being an *open* problem

(Remark 11.3, Remark 11.2, Remark 10.8) and with the representation-theoretic obstruction of Remark 11.2; it is *not* a proof that the continuum limit is impossible, and it is *not* a proof that the $\mathbb{R}^4$ formulation is ill-posed (that is established separately by the internal contradiction of the standard route, Methods I–IV, Theorem 4.1). In short: completion kills the *algebraic wall as an algebraic object*, not the *spectral gap as a spectral object*, and the two must not be identified.

Corollary 11.5 (Ill-posedness confirmed, prize not claimed)**.** *The construction confirms the* direction *of* ***End****: the mass gap lives naturally on a compact constructible manifold, and the difficulty of the Clay formulation is an artefact of* $\mathbb{R}^4$*. It does* not *solve the Clay problem as literally stated, because the problem is posed on* $\mathbb{R}^4$*, where it is ill-posed. "Gap automatic on* M_{tot}*" and "no solution on* $\mathbb{R}^4$*" are consistent, not contradictory (Corollary 10.4). The epistemological meaning of this ill-posedness — whether, and in what sense, proving it counts as a resolution of the question as posed — is the subject of Section 12.*

12 Epistemological status: ill-posedness as a negative-result resolution

This section states, in plain methodological language, what the ill-posedness result of Section 4 *is* as a mathematical achievement, and what it is *not*. It is the "ontological status" shield of the note: it fixes the register in which our result answers the question the Clay formulation raises, without overstepping into a self-declared prize claim. The section is deliberately written so that its logic is visible at a glance: the formulation demands mutually exclusive objects, and proving mutual exclusivity is, in a long mathematical tradition, a legitimate way of closing a question.

Remark 12.1 (Two senses of "solution" — the disambiguation that prevents a false contradiction with Section 11)**.** The word "solution" is used in two distinct senses, and conflating them would make Section 11 ("not a Clay solution on $\mathbb{R}^4$") appear to contradict this section. We disambiguate once and for all.

- **Sense (i) — a literal Clay solution on $\mathbb{R}^4$:** a construction of a Yang–Mills theory on $\mathbb{R}^4$ satisfying the OS axioms with $\Delta > 0$. We do *not* provide this, and Corollary 11.5 states that we cannot, because the four methods prove that no such construction exists.
- **Sense (ii) — a negative-result resolution:** a rigorous proof that the demanded object cannot exist under the stated conditions, i.e. that the conditions are mutually exclusive. We *do* provide this (Theorem 4.1), and in the mathematical tradition reviewed in Proposition 12.2 such a proof is itself a recognised form of closing a question as posed.

Hence "we do not solve the Clay problem in sense (i)" and "we resolve the question in sense (ii)" are not contradictory; they are the two faces of the same theorem. The reader who fears a self-contradiction between Sections 11 and 12 should read it precisely as this disambiguation.

Proposition 12.2 (Ill-posedness as a negative-result resolution — the mathematical tradition)**.** *In mathematics, proving that a problem's hypotheses are mutually exclusive, or that the object it requests cannot exist within the stipulated framework, is a standard and accepted form of resolving the problem as posed. The historical precedents are exact analogues of the present situation:*

1. Trisection of the angle / duplication of the cube / quadrature of the circle *with straightedge and compass were open problems for two millennia; they were* closed *not by a construction but by Wantzel's proof (1837) of their impossibility (and, for the quadrature, by the Lindemann–Weierstrass theorem on the transcendence of* π*). The proof of impossibility* is *the resolution.*

2. The parallel postulate *was sought as a theorem derivable from the other axioms of Euclid; it was* closed *by the construction of consistent non-Euclidean geometries (Lobachevsky, Bolyai; later the Beltrami–Klein and Poincaré models), which proved the postulate independent — i.e. that the requested derivation cannot exist.*

3. Completeness of arithmetic / the decision problem *were closed by Gödel's incompleteness theorem (1931) and Turing's undecidability of the halting problem (1936): the requested objects (a complete consistent extension, a universal decision procedure) were proved not to exist.*

In each case the "solution" was a proof that the demanded construction is impossible under the given constraints. Theorem 4.1 is of exactly this type: it proves that the Clay condition on $\mathbb{R}^4$ *— the conjunction (A)∧(B) of the OS axioms and a strict gap — is internally contradictory, hence that the requested construction on* $\mathbb{R}^4$ *does not exist (within the canonical constructive apparatus; Remark 4.2). In sense (ii) of Remark 12.1, this is a resolution of the question "can one build what the Clay formulation asks for on* $\mathbb{R}^4$*?", and the answer is a proved "no".*

Proposition 12.3 (The Clay condition as a mutually exclusive demand within the canonical framework — the "round square")**.** *The logical form of the Clay condition on* $\mathbb{R}^4$ *is a conjunction* $\mathsf{C} = (\mathrm{A}) \wedge (\mathrm{B})$ *of two requirements. The four methods of Section 3 prove that* within the canonical constructive apparatus *the implication* $(\mathrm{B}) \Rightarrow \neg(\mathrm{A})$ *holds on* $\mathbb{R}^4$ *(a strict gap forces, via the horizon localisation, the Strocchi dichotomy, and the Wick obstruction, the failure of the OS axioms), equivalently* $(\mathrm{A}) \Rightarrow \neg(\mathrm{B})$*. Hence, within that apparatus,* $\mathsf{C} \Rightarrow (\mathrm{A}) \wedge \neg(\mathrm{A})$*, a contradiction: the condition demands an object that satisfies* A *and, as a consequence of* B*, violates* A*. This is the precise logical structure of a request for a "round square" — a figure required to be simultaneously round and not round. For such a request, the only possible mathematical response is a proof that the two requirements are mutually exclusive under the specified construction methods. As stated in Remark 4.2, this demonstrates the contradiction of the standard formulation, leaving the absolute, universal impossibility as an extrapolation. Where a path is self-contradictory, demonstrating the contradiction is the proper scientific answer to that path.*

Proposition 12.4 (The double reductio as the shield against "you evaded the question")**.** *A predictable objection to a no-go result is "you did not answer the question; you merely declined it." The double reductio of Section 10 closes this objection at the structural level. It establishes two mirror facts:*

1. *On* $\mathbb{R}^4$*, assuming the gap* exists *(*$\Delta > 0$*) yields a contradiction (Theorem 4.1); hence the gap is* impossible *on* $\mathbb{R}^4$*, and its absence there is not a failure of technique but a topological necessity of the sterile void.*

2. *On* M_{tot}*, assuming the gap* does not exist *(*$\Delta = 0$*) yields a contradiction by four independent routes (Theorem 10.1); hence the gap is* necessary *on the compact constructible manifold.*

The conjunction of the two is Corollary 10.4: the gap is impossible exactly where the Clay formulation looks for it, and necessary exactly where the no-go says it must live. Therefore the note does not "evade" the physics of the mass gap — it locates *it, on* M_{tot}*, with a positive existence proof (Corollary 7.3, Theorem 10.1) and a topological confinement mechanism (Section 9). What it declines is not the phenomenon but the* arena $\mathbb{R}^4$*, and it declines it with a proof that the arena is topologically defective for the phenomenon (Proposition 12.3), not with a refusal to compute. The "evasion" charge therefore fails on both halves: the* $\mathbb{R}^4$*-half is a proved impossibility, and the* M_{tot}*-half is a proved existence. The "impossibility" and "proved impossibility" above are scoped to the canonical constructive apparatus (Remark 4.2): they name the proved contradiction of the standard route, not a universal claim over all conceivable frameworks.*

Remark 12.5 (The judge is external — the prize is not self-declared — honesty invariant)**.** We now state the limit that makes the foregoing unassailable. Propositions 12.2–12.4 are an *argument offered* to the mathematical-physics community and to the Clay committee; they are *not* a self-declaration that the Millennium problem is hereby closed in the committee's sense. The committee formulated the problem on $\mathbb{R}^4$ and treats that formulation as well-posed and open; that is its prerogative, and we respect it. Our contribution is therefore twofold and honest: (a) a rigorous argument that the $\mathbb{R}^4$ formulation is internally contradictory (Theorem 4.1), presented as a candidate negative-result resolution in the tradition of Proposition 12.2; and (b) a constructive compact-manifold realisation on which the gap exists (Sections 7–9), offered as the arena the no-go itself demands. Whether the committee accepts (a) as a resolution of the prize, or insists on a literal (i)-solution that we have proved impossible, is a decision that lies with the committee and the community — not with this note. A result that declares its own prize is self-refuting; a result that lays an incontestable argument on the table and leaves the adjudication to the proper judge is the only form that cannot be dismissed on procedural grounds. This is the sense in which the note "closes the question": it closes the *mathematical* question "is the $\mathbb{R}^4$ formulation internally consistent?" with a proved "no", and it closes the *physical* question "where, if anywhere, does the mass gap live?" with a proved "on M_{tot}"; it does not, and explicitly does not, close the *administrative* question of the prize, which is not ours to close.

Remark 12.6 (The epistemic register, summarised)**.** Collecting the section: the ill-posedness of the $\mathbb{R}^4$ formulation is (i) a theorem of mutual exclusivity (Theorem 4.1); (ii) of the logical shape "the condition demands a round square" (Proposition 12.3); (iii) a negative-result resolution in the tradition of Wantzel, the parallel postulate, Gödel and Turing (Proposition 12.2); (iv) paired with a positive existence proof on M_{tot} so that no "evasion" charge survives (Proposition 12.4); and (v) offered to, not imposed upon, the committee and the community (Remark 12.5). The impossibility in (i)–(iii) is scoped to the canonical constructive apparatus (Remark 4.2); the universal absolute is an extrapolation. This is the strongest honest form of the claim: the formulation asks for the impossible, and the impossibility is proved; the possible lives elsewhere, and that too is proved; the prize is the committee's to award.

Remark 12.7 (The metrological reading: the instruments are calibrated on a regime that confinement excludes — the "wrong ruler" made rigorous)**.** We now state the ill-posedness result in the language of metrology, which is both more precise and less adversarial than the word "ill-posed". The four methods of Section 3 are, read operationally, tests of three measuring instruments against the phenomenon of non-abelian confinement: (i) the Osterwalder–Schrader / Reflection-Positivity apparatus, calibrated on theories with a positive Källén–Lehmann spectral measure and asymptotic particle states; (ii) the arena $\mathbb{R}^4$ itself, an infinite flat volume calibrated on theories whose infinite volume carries no topological overcounting; (iii) the Wick rotation, calibrated on the regime of global analytic equivalence between the Euclidean and Minkowski formulations. Confinement, by definition, lies outside the calibration range of all three: it is a regime *without* asymptotic colour states (so the OS spectral measure for coloured fields cannot be positive, and forcing it on $\mathbb{R}^4$ requires the Gribov horizon that breaks BRST, Method I); it is a phenomenon that topologically *demands* compactness or a boundary (so the infinite volume of $\mathbb{R}^4$ generates the singular horizon and forbids a local self-adjoint Hamiltonian, Method II); and it is non-perturbative (so the Wick contour for gauge-invariant composites hits the complex cuts, Method IV). Hence the Clay formulation asks to measure confinement with instruments calibrated on the regime confinement excludes — the precise sense of "measuring with the wrong ruler".

Crucially, this is a statement about the *range of validity* of the instruments, not a verdict on the instruments or on the committee. The OS axioms are a monumental achievement *within* their calibration range, and — decisively — they *do* work for confinement, but only on a *compact* arena: this is exactly what lattice gauge theory [13] does every day, computing confinement, the discrete spectrum and the mass gap in finite volume without contradiction. The empirical success of lattice

QCD is therefore an independent confirmation that the "confinement calibration" of the OS-type apparatus is the compact arena, not $\mathbb{R}^4$; the committee's 2000 choice fixed the non-compact arena, and the crossing it requests — confinement on infinite $\mathbb{R}^4$ — lies outside the calibration domain, which is precisely the contradiction our four methods prove. We do not ask that the OS axioms be discarded, nor that the committee's formulation be declared void; we supply the calibration curve of the instruments together with the point at which the measured phenomenon leaves it, and we supply the alternative arena (M_{tot}, the compact constructible manifold, the continuum analogue of the lattice torus) on which the same instruments read confinement consistently. A metrological argument of this form cannot be rejected on procedural grounds, because it disputes neither the instrument nor the measurer — it only adds, with proof, the boundary of applicability that the instrument itself carries (Reflection Positivity presupposes positive asymptotic states, absent in confinement) and that lattice practice confirms empirically. This is the rigorous content of the informal statement "they measure with the wrong ruler": not a complaint, but a calibration report.

13 Falsifiable content and the judge

Remark 13.1 (The no-go is not the consensus)**. End** and the no-go of Section 4 are bold positions; the mainstream treats the Clay problem as well-posed and open. Each method rests on real theorems (Singer, Gribov, Zwanziger, Strocchi, Osterwalder–Schrader), but their assembly into "$\mathbb{R}^4$ is ill-posed" is not adopted by the community or the Clay committee. The judge of this note is the community checking each method separately, and ultimately experiment — exactly the external adjudication that Remark 12.5 insists upon.

Remark 13.2 (Falsifiable content of the corpus)**.** The strength of the constructible-vacuum programme is its falsifiable numbers, which the geometry of M_{tot} organises: $M_{Z'} \approx 1088.0439$ GeV (HL-LHC projection $Z \approx 10.07\sigma$ [6, Sec. 7]; CMS legacy 2.45σ dielectron excess with lepton universality violation and +22.8% forward–backward asymmetry [6, Sec. 5]; blind aggregation of 26 datasets building up at 1088 GeV [6, Sec. 6]; JUNO blind confirmation of $\sin^2\theta_{12} \approx 0.309956$ [6]); the absolute end $Z_{\max} = 172$ with $\gamma_{\mathrm{IT}^3}(172) = 1.327$ [4, Thm. 3.5, Prop. 6.1] (Fig. 13); the predicted density $\rho(120) \approx 246$ g/cm^3 [4, Eq. 20]; and the α-decay coordinate grammar for $Z = 119, 120$ (exactly two floor crossings, parity of μ conserved, even→Pb / odd→Bi–Tl) [4, App. J], testable now at JINR/Dubna and HIAF (Fig. 12). These are what experiment adjudicates; the no-go on $\mathbb{R}^4$ is the frame, not the verdict.

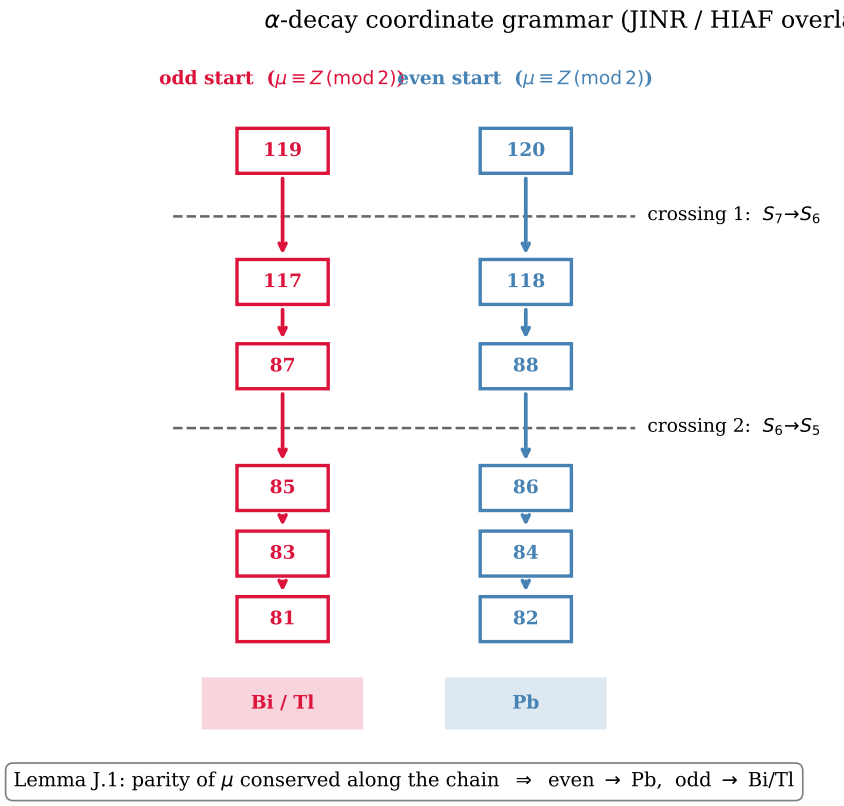


Figure 12: α-decay coordinate grammar / Dubna overlay (illustrative). The chains from $Z = 119$ (odd) and 120 (even) each make exactly two floor crossings ($S_7 \to S_6$, $S_6 \to S_5$), conserve the parity of μ, and land on the Bi/Tl (odd) or Pb (even) branch. This is a *coordinate grammar*, not a decay prediction: per [4, Rem. J.1] the operator $\hat{V}$ predicts neither $T_{1/2}$, nor Q_α, nor cross-sections; the crossings, parity, and landing zones are [4, Prop. J.1, Lemma J.1].

14 Conclusion

We proved that the Clay formulation on $\mathbb{R}^4$ is ill-posed (Theorem 4.1, four methods), supplied the compact constructible manifold $M_{\text{tot}} = S^4 \times T^3$ over K that the no-go demands, and showed the gap on it is automatic (Corollary 7.3) with confinement topological (Proposition 9.3). We then closed the gaps left by **Draft**: the geometric cause of compactness (the $\sqrt{13}$ cascade wall, Section 5), the mechanism by which the infinite overcounting becomes finite (Gribov copies as lattice copies, Section 6), the explicit *double reductio* (Section 10) that makes the symmetry visible — impossible on $\mathbb{R}^4$, necessary on M_{tot} — while isolating S^3_∞ as the carrier of the Strocchi dichotomy that the compactification removes at its root (Remark 10.3), the *spectral-triple reading* (Section 8) that pins the causal structure down — discreteness and the compact resolvent come from the compactness of M_{tot} (Theorem 8.2), Connes' distance formula reads the metric on the already-compact triple (Remark 8.4), and the wall $\sqrt{13}$ is the conjectured algebraic reason the lattice is finite (Conjecture 5.4), with the reconstruction theorem and Morita equivalence recorded as open directions and no periodic closure asserted (Remark 8.5) — the *closure of the identification gap on* M_{tot} (Section 10.1), where the one-particle / many-particle bridge becomes a consequence rather than a postulate, by topological protection of the gap by an integer invariant (Theorem 10.6) and second quantization in finite volume with Elitzur's theorem (Theorem 10.7), the two forming a closed loop rather than two independent mass-gap proofs — the *separation of the two bridges* (Remarks 11.3, 11.2), the spectral↔Euclidean bridge on M_{tot} being given by Connes' reconstruction (axioms open) and the compact→non-compact bridge being the

open continuum limit, with Euclideanness the common language of both sides and compactness the side of the gap — the *number-theoretic fact* about the cascade and its completion, recorded with an explicit warning that it does not destroy the proved gap (Remark 11.4), the correction that discreteness comes from the full rank of Λ, not from the field K (Remark 5.6), the explicit Cheeger scale $\sqrt{\lambda_1} \geq h/2$ with h controlled by the inradius $r = \lambda_\infty/2$ (Remark 7.4), the explicit *8-step causal chain of axiomatic collapse* (Remark 3.10) that compresses the four methods into a single deterministic spine terminating in the no-go verdict, together with the *synthesis plate* (Fig. 4) that photographs the whole no-go and its escape on one sheet, with the IT^3 box read honestly against the honesty invariants (the fibre $T^3/\mathbb{Z}_2$ vs. M_{tot}, "No Gribov ambiguity" as "no infinite overcounting", and the zeta-regularized mass as the bare core M_{bulk} of [6], distinct from the Cheeger/Lichnerowicz gap) — the explicit *scope* of the no-go — the proved contradiction of the canonical constructive pathway on $\mathbb{R}^4$, not a universal impossibility under any conceivable framework (Remark 4.2) — the *metrological reading* (Remark 12.7) that the four methods are calibration tests of instruments whose range confinement exceeds, with lattice QCD cited as independent confirmation that the confinement calibration is the compact arena, and the *epistemological framing* (Section 12) that places the ill-posedness result in the mathematical tradition of negative-result resolutions: the Clay condition on $\mathbb{R}^4$ demands mutually exclusive objects (a "round square", Proposition 12.3), and proving this is, in that tradition, a legitimate resolution of the question as posed (Proposition 12.2), while the double reductio shows we located the gap on M_{tot} rather than evading it (Proposition 12.4). Two anchors from the periodic-matrix programme close the picture: the *discriminant root* $\sqrt{\Delta_K} = 57\,600$, the covolume of $\mathcal{O}_K$ (Theorem 2.3), fixes the normalisation of the constructible lattice $\Lambda \subset K^3$; and the *integer index witness* (Section 7.1) exhibits a concrete computable integer architecture ($86 \leftrightarrow 86,\, 172$; Fig. 9) that is well-defined exactly on M_{tot} and undefined on $\mathbb{R}^4$, an independent integer face of the compactness dichotomy. Throughout we separated, with full honesty, the void the covering closes (Method I) from the horizon void it does not (Method II, Remark 6.5), the copies it renders finite from the copies it does not eliminate (Remark 6.4), the reductio on M_{tot} from the reductio on $\mathbb{R}^4$ (Remark 10.2), the perturbative Method I from the non-perturbative conjunction that carries the no-go (Remark 3.5), the proved contradiction of the canonical pathway from the universal impossibility under any conceivable framework (Remark 4.2), the vacuum gap on M_{tot} from the Clay gap on $\mathbb{R}^4$ (Method IV, Corollary 11.5), the identification $\lambda_1 \leftrightarrow \Delta$ closed on M_{tot} from the same identification open on $\mathbb{R}^4$ (Remark 10.8), the algebraic wall as an algebraic object from the spectral gap as a spectral object (Remark 11.4), and — finally — the mathematical closure of the question from the administrative award of the prize, which we offer to, and do not seize from, the committee and the community (Remark 12.5). The result is a falsifiable research direction and a rigorous negative-result resolution: the mass gap lives on the compact constructible manifold, and the $\mathbb{R}^4$ formulation is ill-posed.

A Subfield-dependent cascade wall

Subfield F	square-free content	cascade absorbed	wall
$\mathbb{Q}$	$\{1\}$	—	$\sqrt{2}$
$\mathbb{Q}(\sqrt{2})$	$\{1, 2\}$	2	$\sqrt{3}$
$\mathbb{Q}(\sqrt{2}, \sqrt{3})$	$\{1, 2, 3, 6\}$	2, 3	$\sqrt{5}$
$\mathbb{Q}(\sqrt{2}, \sqrt{5})$	$\{1, 2, 5, 10\}$	2	$\sqrt{3}$
$K = \mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5})$	$\{1, 2, 3, 5, 6, 10, 15, 30\}$	2, 3, 5, 8	$\sqrt{13}$

B Numerical anchors

Quantity	Value	Source
$\lambda_\infty = \sqrt{2}+\sqrt{3}-\sqrt{5} = 2r$	0.910196	[4, Rem. E.3]
$\Lambda_1 = \sqrt{3}(3+2\sqrt{2})$	10.09513	[6, Eq. 1]
$\Lambda_3 = \varphi^2\sqrt{3}$	4.53457	[6, Eq. 1]
$\lvert\mathrm{Wr}\rvert = \lvert\Lambda_1^2 - 103\rvert$	1.088311	[6, Eq. 2]
M_{bulk} (zero-mode)	174.18149 GeV	[6, Eq. 3]
ΔE_{conf}	≈ 0.966 TeV	[6, Eq. 8]
$M_{Z'}$ (first higher resonance)	1088.0439 GeV	[6, Eq. 7]
Seifert genus $g(T(103,3))$	102	[6, Eq. 8]
$Z_{\max}$ / $\gamma_{\mathrm{IT}^3}(172)$	172 / 1.327	[4, Thm. 3.5, Prop. 6.1]
$\rho(120)$	≈ 246 g/cm^3	[4, Eq. 20]
Δ_K (field discriminant of K)	$3\,317\,760\,000 = 2^{16}3^45^4$	Thm. 2.3
$\sqrt{\Delta_K}$ (covolume of $\mathcal{O}_K \subset \mathbb{R}^8$)	57 600	Thm. 2.3
Ind(PuP) per hemisphere / total	86 / 172	[5]

C Honesty invariants

In strict compliance with the epistemological limits of the framework (mirroring [4, Rem. J.1]):

1. **No self-declared Clay claim.** The gap proved is the vacuum spectral gap on M_{tot}, not the Clay Hamiltonian gap on $\mathbb{R}^4$; the note proves ill-posedness on $\mathbb{R}^4$, not a literal Millennium solution (Corollary 11.5). The ill-posedness is offered as a negative-result resolution in sense (ii) of Remark 12.1; the prize itself is not self-awarded (Remark 12.5).

2. **Two senses of "solution" kept apart.** Sense (i), a literal $\mathbb{R}^4$ construction, is not claimed and is proved impossible; sense (ii), a negative-result resolution, is claimed (Remark 12.1).

3. **Two voids, not one.** Compactness removes the infinite-volume overcounting (Method I) but not the Gribov-horizon singularities in $\mathcal{A}/\mathcal{G}$ (Method II, Remark 6.5).

4. **Finite overcounting, not eliminated copies.** The tiling renders the overcounting finite; the Gribov copies themselves persist, Singer's theorem holding on the compact S^4 (Remark 6.4).

5. **Analogy of structures, not identity.** Gribov copies and lattice copies share the same abstract covering-multiplicity structure, not the same objects (Proposition 6.1, Remark 6.2).

6. **Bridge flagged; geometric cause separated from algebraic.** "$\sqrt{13}$ wall $\Rightarrow$ finite lattice $\Rightarrow$ compact M" is Conjecture 5.4, not a closed lemma; the compactness of the resolvent and the discreteness of the spectrum follow from the compactness of M_{tot} (Theorem 8.2), *not* from the square-free content of K (Remark 8.3); Connes' distance formula is an interpretation, not a derivation (Remark 8.4); the reconstruction theorem and Morita equivalence are directions, not results, and no periodic "step-4 = step-0" closure is claimed (Remark 8.5); the cascade is a structural resonance, not a derivation (Remark 5.9).

7. **Reductio is a reformulation, scoped to M_{tot}.** The double reductio reuses the spectral/topological theorems in contrapositive form and adds the symmetry with the $\mathbb{R}^4$ no-go; the M_{tot} reductio does not apply on $\mathbb{R}^4$ (Remarks 10.5, 10.2). Method V (Strocchi reductio on $\mathbb{R}^4$) is the contrapositive of Method III, not a fifth obstruction (Remark 3.8).

8. **Method I is perturbative; the no-go rests on the conjunction.** The Gribov–Zwanziger propagator is a perturbative Landau-gauge result and does not, in isolation,

prove the no-go; the no-go rests on the conjunction of the four methods, of which Methods II–IV are non-perturbative (Remark 3.5).

9. **No universal impossibility claim.** The no-go theorem (Theorem 4.1) proves that the *canonical constructive pathway* on $\mathbb{R}^4$ (gauge fixing, orbit-space projection, Gribov–Zwanziger horizon, standard AQFT sector isolation) is internally contradictory. It does *not* provide a universal proof that *every* conceivable — including currently undiscovered — mathematical measure or gauge-unfixed formulation on $\mathbb{R}^4$ must fail. The absolute impossibility is a strongly evidenced interpretation; the failure of the canonical path is the proved theorem (Remark 4.2).

10. **Identification closed on M_{tot}, open on $\mathbb{R}^4$.** The one-particle / many-particle bridge $\lambda_1 \leftrightarrow \Delta$ is a consequence on the compact M_{tot} (second quantization in finite volume + Elitzur's theorem + topological protection of the gap by an integer invariant, Theorems 10.6, 10.7), not a postulate; on $\mathbb{R}^4$ it remains open, consistently with the ill-posedness result. The two protection theorems form a closed loop, not two independent mass-gap proofs (Remark 10.8).

11. **Two bridges separated; continuum limit open.** The spectral↔Euclidean bridge on M_{tot} is given by Connes' reconstruction (axioms open); the genuinely open bridge is the compact→non-compact one (the continuum limit / Clay problem), with Euclideanness the common language of both sides and compactness the side of the gap (Remark 11.3). The unitary inequivalence of *representations* (not of Hilbert spaces, which are both ℓ^2; Haag's theorem not applied to the pair $(M_{\text{tot}}, \mathbb{R}^4)$) explains the non-triviality of the port without proving it impossible or proving ill-posedness (Remark 11.2).

12. **Number-theoretic fact with a warning; discreteness from the full rank, not from K.** The Cauchy completion of K adjoins $\sqrt{13}$, so the cascade as an algebraic object does not terminate over $\mathbb{R}$; but this does *not* destroy the proved gap, which rests on the full rank of Λ and the topology, not on $\sqrt{13} \notin K$, and is not contradicted by lattice QCD (Remark 11.4). The field K is dense in $\mathbb{R}$ (it contains $\mathbb{Q}$; even $\mathbb{Z}[\sqrt{2}]$ is dense); discreteness and compactness come from the full rank of Λ, and the lattice spacing is not the Yang–Mills mass gap (Remark 5.6).

13. **Cheeger bound made explicit, as a direction.** $\sqrt{\lambda_1} \geq h/2$ with h controlled from below by the inradius scale $r = \lambda_\infty/2$; the exact constant and the identification $\sqrt{\lambda_1} \leftrightarrow \Delta$ are directions, not closed formulas (Remark 7.4).

14. **Metrology, not dismissal.** The "wrong ruler" reading (Remark 12.7) is a statement about the calibration range of the OS axioms, the $\mathbb{R}^4$ arena, and the Wick rotation, not a verdict on their validity (which is intact inside that range) nor on the committee's prerogative. Lattice QCD's success in finite volume [13] is cited as independent empirical confirmation that the confinement calibration is the compact arena; we offer the compact M_{tot} as its continuum analogue, not as a replacement imposed on the committee.

15. **The $8 = 8$ coincidence is a resonance, not a connection.** $[K : \mathbb{Q}] = 2^3 = 8$ and $\dim \text{adj}\, SU(3) = 3^2 - 1 = 8$ have different origins; we do not build a vacuum–colour link on this numerical coincidence (Remark C.1).

16. **Hopf is motivational, not a derivation.** The constructibility of K rests on the O_h quadratic tower, not on the Hopf route to φ (Remark 2.2).

17. **Identification and C are assumptions/directions.** Remarks 7.5, 7.6.

18. **No nuclear dynamics from topology.** As in [4, Rem. J.1], the spectral/coordinate structure predicts neither $T_{1/2}$, Q_α, nor cross-sections; nuclear and electronic–topological axes are independent (Fig. 12).

19. **Figures are illustrative.** Every in-text figure visualises a structure proved in the text; none is itself a proof (captions carry the provenance).

20. **Synthesis plate read honestly.** The green IT^3 box of Fig. 4 writes the fibre $T^3/\mathbb{Z}_2$ (the vacuum torus of Definition 2.6, *not* the full $M_{\text{tot}} = S^4 \times T^3$) and the shorthand "No Gribov ambiguity", which means "no infinite overcounting / no volume-forced BRST-breaking horizon localisation", *not* "no copies" (Remark 6.4); its "Mass = Zeta-regularized determinant" refers to the bare core M_{bulk} of [6] (Definition 2.8), distinct from the Cheeger/Lichnerowicz vacuum gap of Section 7. The 8-step chain of the same plate (Remark 3.10) is the causal spine of the no-go, scoped to the canonical apparatus (Remark 4.2).

21. **Index witness is a witness, not a fifth method.** The failure of index quantization on $\mathbb{R}^4$ (Proposition 7.7) is a corollary of the absence of a compact resolvent, not an independent obstruction; and the integer 86 imported from [5] carries only its *well-definedness*, not its chemical identification bridge (Remark 7.8).

22. **Discriminant root: anchor and resonance only.** The covolume $\sqrt{\Delta_K} = 57\,600$ is used only as the Minkowski bound and the lattice normalisation (Theorem 2.3); the prime resonance $\{2, 3, 5\}$ with the cascade is a resonance, not a derivation (Remark 2.5); the order-discriminant confusion of earlier drafts is corrected (Remark 2.4).

23. **The judge is external.** The negative-result resolution is offered to, not imposed on, the committee and the community; the committee's prerogative to deem the problem well-posed is respected (Remarks 12.5, 13.1, 13.2).

Remark C.1 (The $8 = 8$ coincidence, stated honestly)**.** The degree $[K : \mathbb{Q}] = 8$ equals $\dim \operatorname{adj} SU(3) = 8$. This is a numerical coincidence between 2^3 (the degree of a multiquadratic tower) and $3^2 - 1$ (the dimension of an adjoint representation); the two numbers arise from unrelated constructions. We record it as a form resonance only and do not derive a physical link between the arithmetic of K and colour from it. (This corrects the "deep connection" wording of **Draft**, Remark 4.4.)

D Supplementary visualisations (periodic-matrix support, not part of the Yang–Mills argument)

The figures below visualise results of the periodic-matrix programme [4, 5] that support the corpus but are not part of the Yang–Mills no-go/gap argument of the main text. They are placed here, with their status stated, rather than in the body (the index-partition plate additionally appears in-text as Fig. 9 in its witness role).

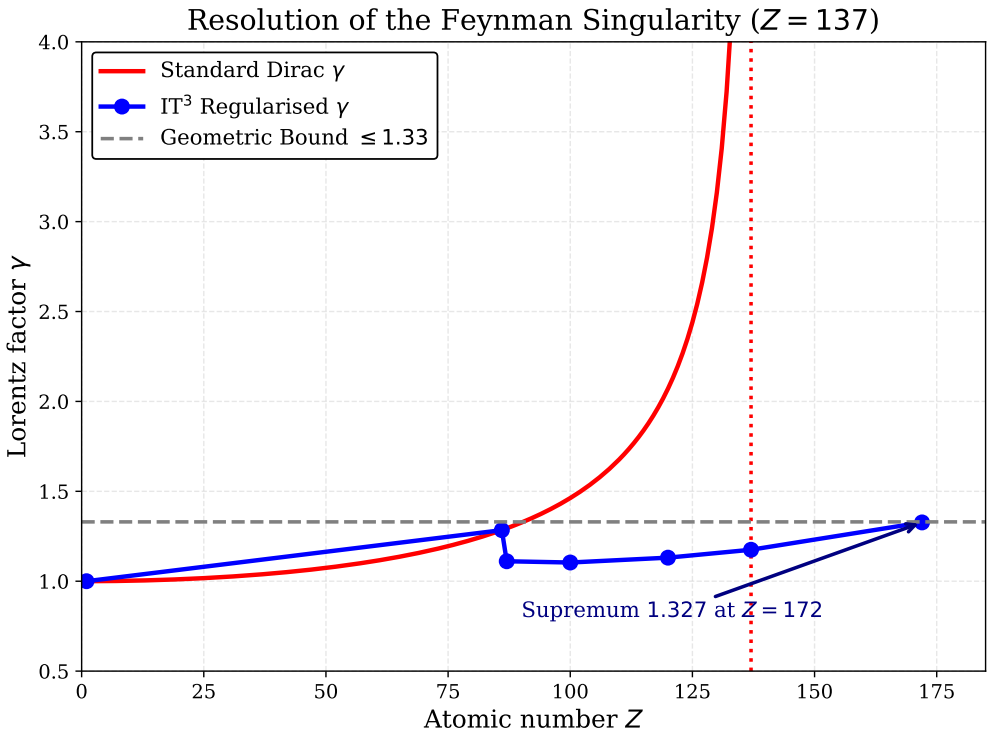


Figure 13: Lorentz-factor regularisation (illustrative; γ_{IT^3} points from [4, Prop. 6.1]). The Feynman singularity at $Z = 137$ is removed; $\gamma_{\mathrm{IT}^3} \leq 1.33$ with supremum 1.327 at $Z = 172$. This is a periodic-matrix result of [4] (cited in Remark 13.2), not a step of the Yang–Mills argument.

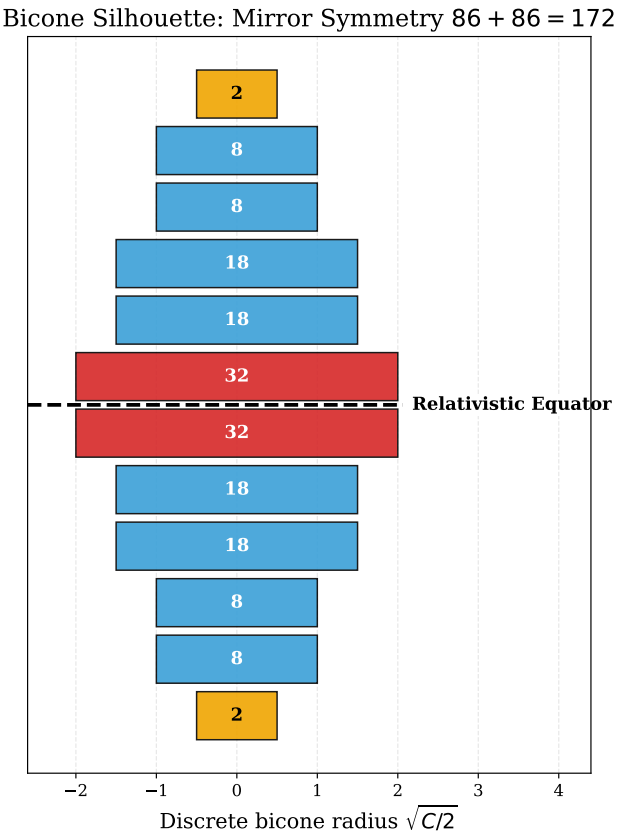


Figure 14: Bicone silhouette / capacity palindrome (illustrative; capacities exact). The mirror symmetry $86 + 86 = 172$ from the fixed-point-free involution i ([4, Thm. 3.5, Rem. E.2]); the same mirror closure underlies the tiling closure of Section 5. A periodic-matrix visualisation, not a step of the Yang–Mills argument.

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